CHAPTER 10

RETRACING AND "RESURVEYING" SECTIONALIZED LANDS

10.1 INTRODUCTION

As described in earlier chapters, the creation of boundaries is unique and demands people who are technically qualified. Yet the retracing of these once-created boundaries also requires a second unique type of person, who probably should be more attuned to the law of evidence than to being a "technology freak." From the beginning, it should be understood in the practice of land surveying that a resurvey and a retracement are not synonymous. The surveyor who elects to conduct an original survey should have different qualifications from those of a retracing surveyor. Retracements require people of different capabilities and possibly specialized knowledge and training.

Chapter 6—Creation and Retracement of GLO Boundaries discussed the creation of the GLO or PLSS boundaries. This chapter discusses the suggested rules for retracing those boundaries. It will discuss retracing metes and bounds boundaries as well. The majority of the modern surveyor's work, except for certain local subdivisions, is to "find the land to be surveyed" by finding the evidence of the original footsteps.

When the words "sectionalized lands" or "public lands" are mentioned, the public lands immediately should come to mind. But the surveyor may find that there are other lands that use that description.

We will address how to find the boundaries to be surveyed and what should be the proper rules and procedures for their retracement. We will concentrate on the rules and methods that should be used to find those original boundaries, with the emphasis on the GLO boundaries, but referencing metes and bounds boundaries as well.

The following principles are discussed in this chapter:

PRINCIPLE 1. An original survey creates original boundaries as well as the evidence of these boundaries. This created evidence and the subsequent written record become the “footsteps” to be followed in a retracement.

PRINCIPLE 2. In any retracement, whether by a federal agency or by a private surveyor, the description contained in the patent of the original boundaries is the controlling description the retracing surveyors should use to place the original boundaries on the ground.

PRINCIPLE 3. When retracing or dividing sections of land created under federal laws, federal rules of survey must be followed (1) when the final adjudication of problems is tried in the federal courts or (2) when a state has accepted or adopted the federal rules and when the original patents were federal in nature.

PRINCIPLE 4. Provided that a superior right is not interfered with or a fraud committed, the boundaries of public lands, when approved, accepted, and patented, are unchangeable and unassailable by individuals or by the courts.

PRINCIPLE 5. Original township, section, and quarter-section corners stand as the true corners that they were intended to represent, regardless of where they are located. This is not true of closing corners.

PRINCIPLE 6. Plats and all original field notes become a part of the grant or patent. When conducting a retracement, the surveyor should consult available and known plats and notes.

PRINCIPLE 7. Closing corners that are not actually located on the line that was closed upon will determine the direction of the closing line but not its legal terminus; the true position is the point of intersection of the two lines.

PRINCIPLE 8. After making allowances for natural changes, a monument, to be identifiable as an original monument, should not differ greatly from the field notes.

PRINCIPLE 9. When there is acceptable evidence indicating the location of an original corner, the corner should be considered as obliterated and should be relocated from that evidence.

PRINCIPLE 10. The original location of a corner may be identified by the evidence as well as by testimony of a witness who saw that corner. It is an exception to the hearsay rule, but the evidence as well as the testimony must be evaluated as having to meet the “preponderance of evidence” criterion, not the “beyond a reasonable doubt” criterion or standard that has been established for criminal matters.

PRINCIPLE 11. In a very limited number of situations, a corner’s position may be located by common use of the monument and point.

PRINCIPLE 12. A corner position may be proven from evidence of found original line trees or from natural features found in close proximity to the corner and identified in the notes.

PRINCIPLE 13. Lost corners may be located by proportioning or by using a combination of evidence and measurements.

PRINCIPLE 14. The surveyor should strive to utilize all forms of evidence to prove a corner as an obliterated corner, rather than resorting to proportional measurements to set a lost corner.

PRINCIPLE 15. When the surveyor has exhausted all possibilities of locating corners of public land surveys from available evidence, only then, as a last resort, should the surveyor rely on measurements alone.

Guiding principles for proportionate measurements are as follows:

A complete discussion of the following principles are listed and discussed in the latest edition of *Manual of Surveying Instructions For the Survey of the Public Lands of the United States*.[1]

PRINCIPLE 15A. When proportionate measurements are used, the new values given to the several parts, as determined by the measurement, should bear the same relation to the record lengths as the new measurement of the whole bears to that record.

PRINCIPLE 15B. A single proportionate measurement should be made between two proven corners, and may locate one or more original points on a line.

PRINCIPLE 15C. Double proportionate measurement usually requires four original corners, and the location of the proportioned corners is positioned at the proportionate distances of lines run from the proportioned points. Primary weight is given to distance, not to angle or bearing.

PRINCIPLE 15D. Lost standard corners will be restored to their original positions on a baseline, standard parallel, or correction line, by single proportionate measurement on the true line connecting the two nearest identified original standard corners. Proper adjustment should be made to secure the correct latitudinal curve. Closing corners are not to be used for either direction or measurement.

PRINCIPLE 15E. When the principal meridian or guide meridian was established by alignment in one direction only, lost township corners on such lines should be restored by single proportionate measurement. Where guide meridians were established as part of the scheme of township boundaries, all surveyed in the same system under a single contract, the township corners located thereon should be relocated by double proportionate measurements.

PRINCIPLE 15F. When lost regular township corners common to four sections and lost regular section corners between township lines were originally established with ties from four directions, the lost corners will be reestablished by double proportionate measurements.

PRINCIPLE 15G. All lost section and quarter-section corners on township boundary lines will be restored by single proportionate measurement between the nearest identified corners on opposite sides of the missing corner, north and south on a meridional line, or east and west on a latitudinal line, after the township corners have been identified or relocated. An exception to this principle will be noted in the case of any exterior whose record shows deflections in alignment between the township corners.

PRINCIPLE 15H. When a line has not been established in one direction from the missing township corner, the record distance will be used to the nearest corner identified in the opposite direction.

PRINCIPLE 15I. Where the intersecting lines have been established in only two directions, the record distances to the nearest identified corners on those two lines will control the position of the temporary points; then, from the latter, cardinal offsets will be made to fix the desired point of intersection.

PRINCIPLE 15J. All lost quarter-section corners on section boundaries within the township will be restored by first establishing or proving the section corners on each side; then the quarter corners will be relocated by single proportionate measurement on line. An exception occurs where the original lines had angular deflection.

PRINCIPLE 15K. Where a line has been terminated with measurement in one direction only, a lost corner will be restored by record bearing and distance, counting from the nearest regular corner, the latter having been duly identified or restored.

PRINCIPLE 15L. A lost closing corner on a standard parallel will be reestablished on the true line that was closed upon, and at the proper proportional interval between the nearest regular corners to the right and left. The only corners that will control the direction of the line being closed upon are (1) standard township, standard section, and standard quarter corners; (2) meander corners terminating the survey of the standard parallel; and (3) closing corners in those cases where they were established originally by measurement along the standard line as points from which to start a survey.

PRINCIPLE 15M. When the north quarter corner on a closing section was originally set and lost, the lost corner will be reestablished on the closing line at a point at the proper proportionate interval between the nearest found or relocated corners to the right and left.

PRINCIPLE 15N. The method to be followed in the subdivision of a section into quarter sections is to run straight lines from the established quarter-section corners to the opposite quarter-section corners; the point of intersection of the lines thus run will be the corner common to the several quarter sections, or the legal center of the section.

PRINCIPLE 15O. Preliminary to the subdivision of quarter sections, the quarter-quarter or sixteenth-section corners will be established at points midway between the section and quarter-section corners, and between the quarter-section corners and the center of the section.

PRINCIPLE 15P. The law provides that where opposite corresponding quarter-section corners have not been or cannot be fixed, the subdivision of section lines shall be ascertained by running lines from the established corners north, south, east, or west, as the case may be, to the watercourse, reservation line, or other boundary of such fractional section, as represented on the official plat.

PRINCIPLE 16. Lost meander corners, originally established on a line projected across the meanderable body of water and marked on both sides, will be relocated by single proportionate measurement after the section or quarter-section corners on the opposite sides of the missing meander corner have been duly identified or relocated. Where a line was terminated with measurement in one direction only, a lost corner will be restored by record bearing and distance, counting from the nearest regular corner, the latter having been duly identified or restored.

PRINCIPLE 17. A meander line platted as the boundary of a riparian owner is seldom the true boundary; the body of water represents the true boundary. Lands acquired by the Swamplands Act and fraudulent surveys go to the meander line.

PRINCIPLE 18. After locating the positions of the meander corners on section lines, the record meander courses and distances are run, setting temporary angle points; the closing error from the last meander course to the section line meander corner is noted. The error of closure is balanced out by the compass rule.

PRINCIPLE 19. Where boundaries that would normally be straight lines were originally established in an irregular manner, the resurvey (retracement) will follow the irregular procedure.

PRINCIPLE 20. Where an original quarter-corner was not originally set, as in the case of sections closing on a correction line or sections closing on a range line with double or triple corners or sections closing on a township line with double or triple corners, place the missing corner on the correction line (or range or township line) at a point between the found or relocated closing section corners a distance that is proportional to the measurements used for the acreage calculations on the original plat. The missing corner is usually set midway between closing section corners except in Section 6, where it is usually 40 chains proportional measure from the northeast or southwest closing section corner.

PRINCIPLE 21. The method to be followed in the subdivision of a section into quarter sections is to run straight lines from the established quarter-section corners to the opposite quarter-section corners; the point of intersection of the lines thus run will be the corner common to the several quarter sections, or the legal center of the section.

PRINCIPLE 22. Preliminary to the subdivision of quarter sections, the quarter-quarter or sixteenth-section corners will be established at points midway between the section and quarter-section corners, and between the quarter-section corners and the center of the section, except on the last half-mile of the lines closing on township boundaries, where they should be placed at 20 chains, proportionate measurement, counting from the regular quarter-section corner. The quarter-quarter or sixteenth-section corners having been established as directed in these principles, the centerlines of the quarter sections will be run straight between opposite corresponding quarter-quarter or sixteenth-section corners on the quarter-section boundaries. The intersection of the lines thus run will determine the legal center of a quarter section.

PRINCIPLE 23. The law provides that where opposite corresponding quarter-section corners have not been or cannot be fixed, the subdivision-of-section lines should be ascertained by running lines from the established corners north, south, east, or west, as the case may be, to the watercourse, reservation line, or other boundary of such fractional section, as represented on the official plat. In this, the law presumes that the section lines are due north and south, or east and west lines, but this is not usually the case. Hence, to carry out the spirit of the law, in running the centerlines through fractional sections, it will be necessary to adopt mean courses, where the section lines are not due cardinals, or to run parallel to the east, south, west, or north boundary of the section, as conditions may require, where there is no opposite section line.

PRINCIPLE 24. Other than creating new subdivisions for private landowners, today the majority of the modern surveyor’s work is retracing the lines of the original surveys, some of which are ancient in nature.

10.2 AREAS OF AUTHORITY

Principle 1. *An original survey creates original boundaries as well as the evidence of these boundaries. This created evidence and the subsequent written record become the “footsteps” to be followed in a retracement.*

The term *survey* may have different meanings to different people. A surveyor is usually asked, “Did you conduct a survey?” The response, usually is yes or no. The surveyor could reply, “Do you mean the noun, the verb, or the adjective ?” Surveying as a generic term may take on multiple meanings, depending on the circumstances and whomever you ask.

In conducting retracements and resurveys, two areas of authority may be found. The ultimate authority in a resurvey of public lands is the Bureau of Land Management (BLM), the administrative successor to the General Land Office (GLO), but through the authority of the U.S. secretary of the interior. The ultimate authority in a retracement by a private surveyor lies in the authority granted to the registered surveyor by the registration agency of the respective state. Technically and legally, no private registered surveyor may conduct a *resurvey* of a parcel that is described in reference to a federal survey by an aliquot parcel description. Surveyors have authority only to conduct retracements, which in and of themselves are binding on no landowner or other surveyor, whereas a retracement should be considered nothing more than an opinion of the corners and lines based on the evidence of the original survey that is recovered. This can include tangible evidence as well as measurements, both original and subsequent.

The Bureau of Land Management is the only agency that Congress has authorized to conduct original surveys of the public domain land and is also authorized to promulgate rules and regulations to put into effect the land laws of the United States. These rules and guidelines apply only to the agency that created the rules. This agency issued various manuals and a pamphlet entitled *Restoration of Lost and Obliterated Corners*. The pamphlet summarizes the findings of federal courts on how to make resurveys and how to divide protracted portions of sections. The information contained in this pamphlet is also included in one of the *Manuals*. Although the instructions given on how to make original surveys are entirely within the authority of the BLM, the rules for resurvey are acceptable only when they agree with court findings. As noted, the pamphlet summarizes federal court findings, but not those of all state courts. Also, because it does not contain instructions on how to resurvey in special situations, other sources of advisory information must be sought.

In this context, no state legislature can enact survey laws for the retracing or reestablishment of public land survey corners or lines that are in conflict with the federal laws under which the original lines were surveyed or created. This also applies to the retracing of federally created corners and resurveying and replacing lost corners.

Principle 2. *In any retracement, whether by a federal agency or by a private surveyor the description contained in the patent of the original boundaries is the controlling description the retracing surveyors should use to place the original boundaries on the ground.*

10.3 RESURVEY OR RETRACEMENT

A few states have adopted the *BLM Manual* and the *Manual*'s definition of resurvey; thus, in those states it is binding. The *Manual* reads: "A resurvey is a reconstruction of land boundaries and subdivisions accomplished by rerunning and re-marking the lines represented in the field-note record or on the plat of a previous official survey. The field-note record of the resurvey includes a description of the technical manner in which the resurvey was made, full reference to recovered evidence of the previous survey or surveys, and a complete description of the work performed and monuments established. The resurvey, like an original survey, is subject to approval of the directing agency.”

As discussed in an earlier chapter, an *original survey* creates boundaries, and without an original survey there would be no boundaries. But once the boundaries are created, no lesser survey agency has authority to modify or alter the original boundaries, other than the agency that created them; and that agency has no authority to change these boundaries if *bona fide rights* granted previously are in any way impinged upon.

Since an original survey can never be re-created, these agencies and registered surveyors are authorized to conduct other classes of surveys. For other states, the definition of *resurvey* is suggestive only, in that resurveys are classified into two major categories. First we should consider a *retracement survey* or *retracement*.

Once an original survey is conducted and accepted, that parcel can never be resurveyed. It can only be retraced and the original measurements defined to more modern or precise references.

According to the 1973 *Manual*, a retracement merely remeasures lines and identifies monuments or other marks of the prior established survey that is being retraced, as well as recovering any uncalled-for monuments, without the restoration of lost corners or reblazing of the original lines. Perhaps the best definition would be to consider a *retracement* as merely an *investigation survey* to determine which original lines and corners can be found and identified. In public land states, the authority of the originally creating authority was first the U.S. Treasury Department, then the General Land Office, and now the Bureau of Land Management. In metes and bounds states, case law decisions, as well as the various definitions of land surveying, become the critical criteria. In a *retracement*, the surveyor attempts to recover as much as possible of the original survey evidence of the original lines. Then, if the surveyor has been given the authority by the original creating agency or its successor, the surveyor surveys and locates the lost corners based on the accepted principles outlined and described in the various manuals. But in metes and bounds states, no such manuals exist, and the numerous *minimum technical standards* that surveyors should understand and adhere to mention only the technical aspects of the works, not the legal aspects. In other words, the minimum technical standards state that the surveyor should close surveys to a specific standard, but *absolutely no* technical standard states that the retracing surveyor has to be on the correct corner.

10.4 TYPES OF SURVEYS AND RESURVEYS

Although a private surveyor does not conduct resurveys of lands that were once public lands, it is important that all surveyors understand the distinctions among the various terms that concern surveys and resurveys.

These may be listed in descending order of control. The recognized classes are as follows:

I. Original surveys
II. Retracements
III. Resurveys
  A. Dependent resurveys
  B. Independent resurveys

First is the original survey. The law holds that the original survey does not describe boundaries, but it creates the boundaries. That *original* survey creates evidence and leaves this evidence behind for the subsequent surveyor to find when a *retracement* is conducted. Probably the best definition of a retracement can be found in the Florida decision *Rivers* v. *Lozeau*, 539 So. 2d 1147 (Fla. App. 1989), when the court wrote:

> [A] surveyor can be retained to locate on the ground a boundary line which has theretofore been established. When he does this, he "*traces the footsteps*" of the "original" surveyor in locating evidence of the boundaries. Correctly stated, this is a "*retracement*" survey, not a resurvey, and in performing this function, the second and each succeeding surveyor is a "following" or "tracing" surveyor and his sole duty, function and power is to locate on the ground the boundaries, corners and boundary line or lines established by the original survey. He cannot establish a new corner or new terminal point, nor may he correct errors of the original surveyor. He must only track the footsteps of the original surveyor. The following surveyor, rather than being the creator of the boundary line, is only its discoverer and is only that when he correctly locates it. (Emphasis added.)

A retracement is the intervening step between the original survey and the resurvey, because this retracement is conducted to recover the evidence created and left by the original surveyor. Then, predicated on the recovery of the evidence in the retracement, a decision is made about conducting either a *dependent* resurvey or an *independent* resurvey.

A *dependent resurvey* is first a retracement of all recoverable evidence of the original corners and lines and then a reestablishment of lost or obliterated corners and lines in accordance with the proper rules, and then in accordance with the best available evidence and applicable rules of survey. The framework of the new survey is dependent on the recovery of the original corners and evidence of any lines. An *independent resurvey* casts aside the original survey and all of its boundary lines and creates all new monuments and corners and may include the establishment of new township lines without reference to the original survey.

During an independent resurvey of a township, parties with bona fide rights usually have their land delineated by a metes and bounds survey and designated by a tract number. In southern California, near the Mexican border, several townships have numerous tracts platted and surveyed without showing a relationship to the original survey. Also, as these tracts are shown with bearings of north, south, east, or west, and distances are in even multiples of 20 chains, the tracts could not possibly be based on the original survey. Without a corrective deed (patentee deeding the patent back to the government and the government deeding the tract to the patentee), title companies refuse to insure title by the tract number. In effect, this stops the development of new subdivisions because title insurance is not available.

In theory, once an independent resurvey is approved, the original survey no longer exists even though monumentation of the original survey may still exist. This is true for land always owned by the federal government, but not for patented parcels existing prior to the independent resurvey. In one township in California (Blaney Meadows), the government decided that the original survey was fraudulent and ordered an independent resurvey. During the independent resurvey, the old original corners were destroyed (although tied out), and the bona fide patent owner was given a new location. As the owner gained several hot springs, he did not complain.

The landmark retracement case, *Cragin* v. *Powell*,[2] contains a reference to a letter from the commissioner of the General Land Office, which reads:

> The making of resurveys or corrective surveys of townships once proclaimed for sale is always at the hazard of interfering with private rights, and thereby introducing new complications. A resurvey, properly considered, is but a retracing, with a view to determine and establish lines and boundaries of the original survey . . . but this principle of retracing has been frequently departed from, where a resurvey (so called) has been made and new lines and boundaries have often been introduced, mischievously conflicting with the old, and thereby affecting the areas of tracts which the United States had previously sold.

10.5 COURT OF PROPER JURISDICTION

Principle 3. *When retracing or dividing sections of land created under federal laws, federal rules of survey must be followed (1) when the final adjudication of problems is tried in the federal courts or (2) when a state has accepted or adopted the federal rules and when the original patents were federal in nature.*

A surveyor retracing a federally created township is bound to follow the federal rules of survey that created that township. This may be a problem if the surveyor determines that the respective state has enacted legislation or follows rules that are different. Pragmatically, if property rights are created under one set of surveying rules, it hardly seems reasonable that a second set of surveying rules should be followed to relocate the lines. A young surveyor may encounter “old-time surveyors” who either refuse to accept this principle or do not understand it.

Once land passes from the federal government to private parties within a state, the jurisdiction over court trials passes from the federal courts to the state courts. Although most state courts recognize federal court rules as applicable in resurveys, this is not universally true. For example, the Missouri state court accepts that state’s statute regarding restoration of a lost section corner, which is quite different from the requirements of federal courts; and in California courts, topography calls have much greater force than that given in federal courts.

10.6 FEDERAL PATENTS

A federal patent is the legal means by which the federal government grants lands from federal ownership into private ownership. A patent carries no warranty of title or any other warranties, so for all practical and legal purposes it may be equated to a quitclaim deed. For the most part, a patent acts as a quitclaim deed granted from the government to the individual. A patent from the federal government is senior to all other rights except prior-granted rights by some previously authorized entity. It is unusual, but not impossible, to determine that several patents were granted to the same parcel. In these instances, the senior patent must be determined, and it will control. When conducting research for a retracement, a surveyor will find it advisable and probably necessary to extract the title to the original patent to be certain to determine what is to be surveyed.

When conducting a retracement, it may be advisable that the surveyor "take" the description back to the original patent. Because of the numerous and varied ways that patents were granted to individuals, states, and corporations, it should be realized by the retracing surveyor that a patent may be granted to a state by one description, yet the state may convey the same land using a different description, thus having different boundaries to be retraced.

10.7 INTENT OF THE GOVERNMENT

In law, the actual proven intentions of the parties to a land transaction are paramount to other considerations, senior rights excepted. Because official government subdivisions were conducted and made under the statutes in effect at the time of surveys and because it is presumed that the instructions, rules, and regulations of the General Land Office were obeyed, any intent on the part of the government must be interpreted from the statutes, rules, and regulations in effect as of the date of the patent. By federal statute, "the boundary lines actually run and marked in the surveys returned by the surveyor general shall be established as the proper boundary lines of the sections or subdivisions for which they were intended." Thus, the original lines as run by the original surveyors are the best indicators of intent. It becomes axiomatic that because the retracement depends on the original surveys described by the plats and field notes, the surveyor who undertakes any retracement must base the retracement survey on the original notes and plats by which the descriptions were created.

10.8 SENIOR RIGHTS

From the inception of the land distribution program and its surveys, the federal laws never made provisions for the federal government to convey prior land rights that were not obtained. The federal land system was one of the few systems providing for recognition of valid land rights that were granted in areas of the public domain prior to the American Revolution. French, Spanish, some Dutch, and even English land grants were evaluated and decisions were made as to their validity. There were several instances in which the validity of the grants was determined after the public land survey network was surveyed on the ground. In some instances, this placed a senior survey (GLO) over a senior grant (e.g., a Spanish grant). There were many instances in which the senior grants existed only on paper and not on the ground.

When a surveyor is asked to retrace either a township, with its sections, or a prior senior grant, the surveyor must examine the date of the respective surveys to determine which is senior or junior as well as the date of the grants.

It is presumed that when a land grant is superimposed within a township, it is senior in title, but it may be junior in the survey of its defining lines.

Provisions were never made for the federal government to sell or convey valid land rights acquired by private citizens prior to establishment of the public domain. A government survey encroaching upon a private grant, such as a rancho land grant, cannot control for conveyance purposes, but the set monuments can be used as reference monuments. In encroached areas, all rights originating from the public domain cease at the boundary of the grant closed upon.

10.9 FOLLOWING THE FOOTSTEPS

Any surveyor who conducts an original survey or an independent resurvey (GLO) is the person who creates the footsteps to be followed in a subsequent retracement. The retracing surveyor (private) follows the footsteps of the creating surveyor. However, this term has presented difficulty in application and in interpretation. If one were to follow the dictum to the "letter of the law," it would mean that the retracing surveyor should first run the section lines with a compass, or solar compass or solar transit, and then use a 2-pole chain, or a 5-chain ribbon tape, to conduct the initial retracement.

Following the footsteps simply means following the *evidence* of the footsteps, which includes the physical evidence as well as methodologies of how the work was accomplished.

Courts and surveyors sometimes rely on the advice to "follow the footsteps of the original surveyor" or, as several courts have written, "follow the footprints," to justify their findings and, in many instances, attempt to discredit the work of other surveyors. No one can walk in the exact steps of the original surveyor; rather, it becomes a matter of "tracking the footsteps" by the evidence that is recovered, and then being able to say with a great degree of certainty, "This is where the surveyor walked."

A surveyor in private practice who is testifying in court will probably be asked, "Did you follow the footsteps?" This question has been asked by attorneys in both federal and state courts. The answer should be "No." No two surveyors can create the same footsteps because the "footsteps" are in reality the evidence of the original surveys. The retracing surveyor should attempt to recover the *totality* of the evidence of the original surveyor and then determine where the original surveyor stood. A reply should be: "I found the evidence of the original footsteps." The evidence should include the instructions, the field notes, the plats, and any other evidence, both documentary and physical, that can be found at the time of the retracement. To "follow the footsteps" implicitly would require using the same equipment, using the same methods for computations, and using the same methods of calculating the areas of the respective sections.

10.10 LINES MARKED AND SURVEYED

Principle 4. *Provided that a superior right is not interfered with or a fraud committed, the boundaries of public lands, when approved, accepted, and patented, are unchangeable and unassailable by individuals or by the courts.*

Congress intended that subsequent surveyors and all courts be restricted or limited in their treatment of public lands relative to inputting their own opinions as to whether they should be retraced. The foundation of the public land system and resurvey procedures depends on this principle, first approved by the courts and later enacted by statute:

> The boundary lines, actually run and marked in the surveys returned by the surveyor general, shall be established as the proper boundary lines of the sections and subdivisions for which they were intended.[3]

The burden of proof is now placed on the retracement surveyor to locate the lines as they were actually placed on the ground, not where they should have been. The resurvey becomes a question of evidence and the ability to convince the judge and/or the jury that the evidence presented is superior to all other evidence collected and presented.

10.11 ORIGINAL CORNERS

Principle 4. *Provided that a superior right is not interfered with or a fraud committed, the boundaries of public lands, when approved, accepted, and patented, are unchangeable and unassailable by individuals or by the courts.*

In many instances, one may find multiple plats for the same township, with each plat depicting different and conflicting information. A federal township plat does not become effective until it is signed by the surveyor general. Along with this principle the footsteps of the township may not be readily identified in that the retracing surveyor must determine which one of the plats was used to identify the parcels that were patented.

The preceding principle is repeated because of its importance. When doing retracements in public lands the federal survey laws apply. There have been instances when states have attempted to enact guidelines that circumvent the federal laws, but these have proven to be inapplicable.

Principle 5. *Original township, section, and quarter-section corners stand as the true corners that they were intended to represent, regardless of where they are located. This is not true of closing corners.*

As discussed earlier, in the Public Land Survey System, the original corners control. Yet in almost all instances, those corners were identified by set monuments and their record evidence. This basic principle was identified by Congress on February 11, 1805, in the land act named for that date. This is a logical sequence of the following principle:

> The boundaries of the public lands, when approved and accepted, are unchangeable and unassailable through the courts.
> The monuments set at the time of the original survey are the best evidence as to where the original boundaries were established; as such, the monuments must remain unchangeable. Closing corners are an exception to previous principles.

10.12 ORIGINAL FIELD NOTES AND PLATS

Principle 6. *Plats and all original field notes become a part of the grant or patent. When conducting a retracement, the surveyor should consult available and known plats and notes.*

In a now famous U.S. Supreme Court decision, this principle was stated as follows:

> It is a well settled principle that when lands are granted according to an official plat of the survey of such lands, the plat, itself, with all its notes, lines, descriptions and landmarks, becomes as much a part of the grant or deed by which they were conveyed and controls so far as limits are concerned, as if such descriptive features were written out upon the face of the deed or the grant itself.[4]

Government field notes are the major source from which the descriptive information concerning corners, monuments, accessories, and line information can be found. Rarely will information concerning witness trees and natural objects appear on the face of the plat. The best evidence of how and where the lines were run appears in the notes. If there is a discrepancy of measurements between the plat and field notes, the plat usually must give way to the field notes, and the land department may properly correct the plat so as to conform to the field notes.[5] However, if a discrepancy affects the description, the plat controls.

For example, during a dependent survey, the original quarter corner, though out of position, was identified and called the quarter corner in the field notes. However, the drafter decided when platting the notes that the section would be skewed and, thus, renamed the original quarter corner the "witness corner," identifying the quarter corner at midpoint. Because the plat is what was patented, the plat controls.

Field notes should be used in conjunction with the plats, because they may explain why certain anomalies are on the plat. See Figure 10.1.

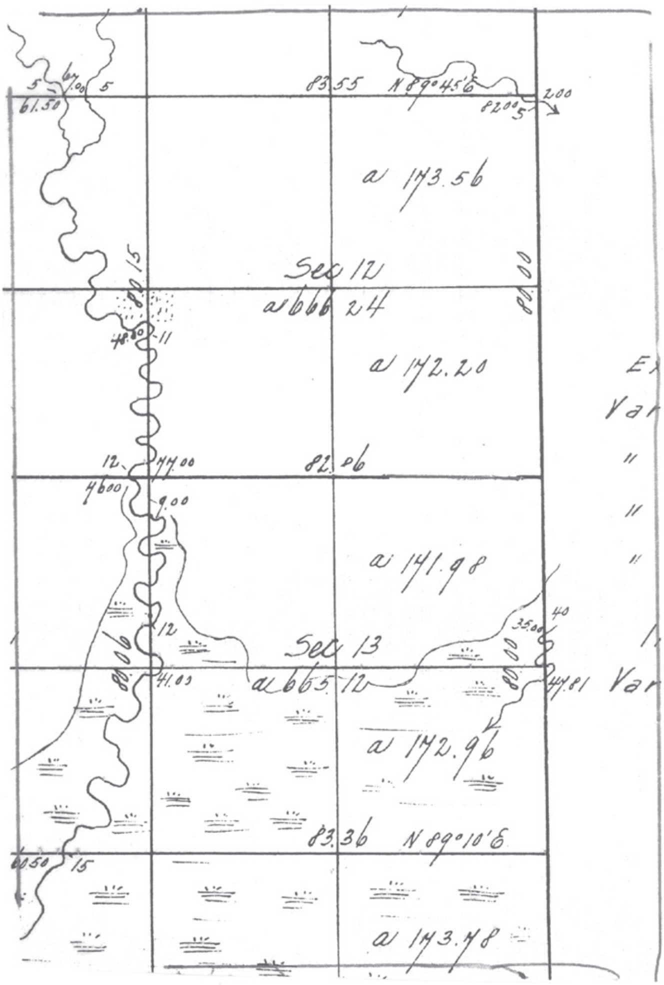


Figure 10.1 Portion of township plat in Mississippi. *Note the following lines:* 1. North line of Section 1—East and West although not indicated. Field 2. North line of Section 12 N 89 45 E 3. North line Section 13 no reference. What is it. Field notes not consulted 4. South line Section 13, N 89 10 E.

The problem: The surveyor did not understand how the variance in bearing occurred. The surveyor took the mean of 2 and 4 and applied the mean to 3, not realizing that the field notes showed it to be east and west. He indicated a mean bearing of N 89 27 W.

The solution: Had the surveyor consulted the field notes, he would have seen in both situations that when the closing lines were run in from the west the field notes indicated a closure or "falling" of "35 links North of the originally set corner. Now multiply 35 by 3 and divide by 7 and the result is the number of minutes of correction = 15 minutes correction, this 89 45 to the North line of Sec. 12.

10.13 CLOSING CORNERS

Principle 7. *Closing corners that are not actually located on the line that was closed upon will determine the direction of the closing line but not its legal terminus; the true position is the point of intersection of the two lines.*

A surveyor who is asked to retrace the lines of those sections that border the range lines or the township lines will encounter closing corners. It should be understood that a closing corner, even though it is a corner, should not be considered as having the same dignity as that of a standard section corner. Because these corners were set after the original township and range lines were created, the closing corner is important but not absolutely controlling. In the running of the closing corner, the surveyor intended that these corners be set on line between two original corners, when in many instances they were simply "estimated" by the original surveyor when he or she ran the original section lines.

Once a line is run and established, it cannot be altered at a later date. If the surveyor failed to place the closing corner on the true line closed upon, the original line cannot be changed to fit the new corner; the closing corner must be moved to the line closed upon. This rule applies to closing corners on standard parallels, correction lines, land grant lines, and the last-set double corner. When a new township is being surveyed or an old one being resurveyed and double corners are set along the township lines, the second corner set must be considered a closing corner.

10.14 IDENTIFICATION OF CORNERS AND LINES

Any surveyor who undertakes to conduct a retracement or what some may call a resurvey must be trained in the area of evidence recovery and evaluation. A retracement or resurvey does not require the services or capabilities of a top technical expert, but, rather, a retracement requires the services of a person who understands evidence, including its significance and its legal implications. Evidence of the original surveys and their significance, and how this evidence relates to the original measurements as well as to modern measurements and not to technical aspects of modern surveying, is the most important element of a resurvey or a retracement.

A surveyor will appreciate that evidence can consist of verbal and real evidence, as well as hearsay evidence. All are equally important in determining where the original footsteps were left.

10.15 MONUMENTS AND THEIR IDENTIFICATION

Principle 8. *After making allowances for natural changes, a monument, to be identifiable as an original monument, should not differ greatly from the field notes.*

All surveyors should realize that monuments set at the corner points are like people. Time affects each differently and in its own way. (1) The character and dimensions of the monument in evidence should not be widely different from the record; (2) the markings in evidence should not be inconsistent with the record; and (3) the nature of the accessories in evidence, including size, position, and markings, should not be greatly at variance with the record.[6]

Before any attempts are made to conduct a retracement, the original field notes of surveys must be obtained from the proper authorities. If more than one "original survey" is available, it should be determined which are applicable by examining court records or dates of survey. Next, a complete chronology of monuments, descriptions, accessories, line objects, and calculations is made.

A variety of monuments have been used throughout the years; posts, stones, stone mounds, and dirt mounds with pits were most frequently used. When undisturbed, stones or stone mounds last for decades, and, on occasion, portions of wooden posts have been uncovered after 100 years. In one area, wooden posts, established in 1799, were recovered in the l950s.

The type of monument found must correspond in size, species, and general character to that called for in the field notes. Deviations from the called-for information make the acceptance of location difficult, and a completely different call makes the corner suspicious. In one area, the original survey called for "wooden posts"; however, today each corner recovered is monumented by a marble post of unknown origin. Here, reputation and common usage come into play—is there better evidence?

Some of the problems that a retracing surveyor may encounter in evaluating evidence recited in field notes as to what is found on the ground may be a variation of materials or variations of distances, bearings, or other recorded evidence. For example, the field notes call for a sandstone monument, 6 by 12 inches, but the surveyor recovers a basaltic monument the same size and marked "as called for." Are these the same? Or a white oak post, 4 inches square marked with two notches is called for, but what is recovered is a red oak post, marked in the same manner. Is it the same post? Or the notes may refer to a bearing tree as being "12 Cyp., N 23 E, .33 ch.," but the surveyor finds a 23-inch cypress, with an old face blaze facing the corner point, whose bearing is S 22--½ W, 34 links. Is this the same tree? Is this the corner point?

By reading the signs of natural wearing and aging, such as the buildup of moss on stone or wooden corners, the approximate time of placement can be determined.

Each stone in a mound is removed and examined to determine the nature of the ground under it and the presence of undecayed leaves. A change in coloration at ground level indicates aging.

The frailties of the creating surveyor become important. Did the creating surveyor estimate the size of the stone, or did he measure it? Did the creating surveyor read the wrong end of the compass? Did the creating surveyor measure to the bearing tree or “guesstimate” it? These are questions that will go unanswered forever. Yet the retracing surveyor must address these questions before an opinion can be formulated.

Few original posts or dirt mounds are recovered because of rotting, fire, or weathering. Redwood and red cedar are naturally long lasting, and white oak is more durable than red oak. “Lighter” pine has a large concentration of resin, which resists rotting but allows the wood to burn readily. In prairie areas, the pits and mounds have long since disappeared, yet the telltale signs of changes in vegetation, soil color, or texture help the surveyor to determine where the pits and mounds were once located.

Accessories such as bearing trees called for in the writings have equal dignity with the corner itself. Blazed bearing trees can be traced back to the time of the original survey by counting tree ring overgrowth. Because trees always grow larger, the reported size of the tree at the time of the original survey can be compared with its present size. In a few locations of very poor soil, however, growth may be slight, and some palm trees remain the same size. Identification of stumps to determine the species is also sometimes important. Bearing trees were usually scribed, and the scribing should be consistent with the field notes or the standard instruction of the time. If the notes for Section 26 call for the scribing to be in Arabic and in the resurvey Roman numerals XXVI are found, a problem arises. On the other hand, if the notes call for a “BT” on each bearing tree and if during a retracement of the entire township no “BT” is found on any tree, is this an error or an omission? Here, the surveyor must rely on his or her judgment to make a decision.

10.16 EVIDENCE OF CORNERS

Principle 9. *When there is acceptable evidence indicating the location of an original corner, the corner should be considered as obliterated and should be relocated from that evidence.*

A surveyor should never hesitate to consider every form or type of evidence to locate a corner monument. Because a corner is the controlling point for a line, courts and surveyors should understand that no form of evidence should be cast aside that could help to locate the original position. Courts consider a lost corner so serious that any shred of acceptable evidence may be utilized to help raise the dignity of a lost corner to an obliterated corner. Before a corner can be considered as lost, the surveyor must research all available records from such varied sources as utility companies, title companies, highway departments, private surveyors, and public agencies for any supporting data. In many western states, the railroads were built very soon after or at the same time as the original surveys were conducted. The right-of-way drawings depict many tie-outs to original section and quarter corners as related to mile posts.

Courts look at the standard of evidence needed to accept corners quite differently than *does* the surveyor and particularly the BLM. Whereas the BLM once required that the degree of proof be *beyond a reasonable doubt*, which is the most stringent, the courts as well as the new edition of the *BLM Manual* now require only the *preponderance of evidence* or creditable evidence.

10.17 USE OF TESTIMONY IN BOUNDARIES

Principle 10. *The original location of a corner may be identified by the evidence as well as by testimony of a witness who saw that corner. It is an exception to the hearsay rule, but the evidence as well as the testimony must be evaluated as having to meet the "preponderance of evidence" criterion, not the "beyond a reasonable doubt" criterion or standard that has been established for criminal matters.*

Courts and the *Manual* accept that testimony of eyewitnesses to the original corners can be critical in determining the location of original corners. Yet they accept the fact that some people have *selective memory* or *selective recall* of facts. They also accept that some people may be prone to be less than truthful, even under oath. Generally, there is a modern trend, when accepting testimony, to identify the location of a corner position to use the "preponderance of the evidence" standard or requirement and not the "beyond a reasonable doubt" standard. Several states have addressed this, and the courts have universally held that *no* agency can set or establish a legal standard for the acceptance of evidence. Testimony is a form of evidence that is important in determining facts. When testimony is used it must meet all requirements relative to the individual who is testifying and the presentation of the evidence.

The Federal Rules of Evidence [Rule 803(20). Hearsay Exceptions] and many state rules of evidence or statutes give special consideration to ancient testimony. In Hawaii, ancient testimony has been given the name *kamaaina*. The Hawaiian court in Pulehunui, used *kamaaina* testimony to determine the boundaries of Pulehunui, a central Maui grant of 16,000 acres that was granted by name only.[7] *Kamaaina* was described as "a person familiar from childhood with any locality." Ancient testimony has been applied to land, evidence, and events.

When a surveyor receives testimony while in the field, he or she should consider the applicability of the field-related testimony to future use. Several states permit surveyors to take oaths for the establishment of old corners and recognized lines. The California law provides:

> Every licensed land surveyor or registered civil engineer may administer and certify oaths: *(a)* When it becomes necessary to take testimony for the identification or establishment of old, lost or obliterated corners. *(b)* When a corner or monument is found in a perishable condition and it appears desirable that evidence concerning it be perpetuated. *(c)* When the importance of the survey makes it desirable to administer an oath to his assistants for the faithful performance of their duty. A record of oaths shall be

> preserved as part of the field notes of the survey and a memorandum of them shall be made on the record of survey filed under this article.

To be usable in the future, statements made under oath must contain facts that are the personal knowledge of the person giving the testimony. Statements of facts about what a person was told by third parties is considered hearsay evidence or double hearsay, or, as some courts have called it, hearsay on hearsay. A witness can testify that he or she personally observed a corner being destroyed when the county road was widened, and he or she can point out to the surveyor the location of the monument as he or she remembers it, but he or she cannot testify where a neighbor said it was located.

A surveyor should place little reliance on the testimony of people who are involved in the land dispute or on statements about boundaries or corners that originated after the dispute began. The *BLM 1973 Manual* (p. 130) describes reliable testimony as follows:

> A corner will not be considered as lost if its position can be recovered satisfactorily by means of the testimony and acts of witnesses having positive knowledge of the precise location of the original monument. The expert testimony of surveyors who may have identified the original monument prior to its destruction and thereupon recorded new accessories or connections, etc., is by far the most reliable, though land owners are often able to furnish valuable testimony.
>
> The testimony of individuals may relate to the original monument or the accessories, prior to their destruction, or to any other marks fixing the locus of the original survey. Weight will be given such testimony according to its completeness, its agreement with the original field notes, and the steps taken to preserve the original marks. Such evidence must be tested by relating it to known original corners and other calls of the original field notes, particularly to line trees, blazed lines, and items of topography.

10.18 COMMON USAGE

Principle 11. *In a very limited number of situations, a corner’s position may be located by common use of the monument and point.*

The Federal Rules of Evidence, as well as most state rules of evidence, accept the fact that boundaries can be identified by long common usage. Several of the courts have extended this exception to include the termini of these lines, the corners, as being determinable by common usage or as some courts call *repose*. In many areas, state and local highways were initially located along section lines with full knowledge of the individual landowners and local residents. Where, by local acceptance, such roads have been superimposed upon the section lines by common usage, local surveyors accept these points as being the best remaining evidence. In the absence of better or more conclusive evidence, a section corner that falls within road intersections may be considered as obliterated. Until recently, this privilege has not been extended to permitting people to testify as to ownership. But unfortunately, several state courts have extended this exception to the hearsay rule to permit a person to testify as to who they thought owned the land.

This certainly flies in the face of historic English law that requires a written document to prove one's claim to title.

When courts rely on the acceptance of a period of "long" prior usage to justify the acceptance of a corner as being the "true" corner without the supporting physical evidence, the courts usually justify their action by calling it a *corner of repose*.

10.19 USING RECORDED INFORMATION TO LOCATE ORIGINAL LINES

Principle 12. *A corner position may be proven from evidence of found original line trees or from natural features found in close proximity to the corner and identified in the notes.*

The original surveyors, who, for the most part, were meticulous and professional, kept excellent field notes. Many of the original notes call for marked line trees, natural features, and numerous other identifiable features. If the retracing surveyor is able to ascertain these original trees and other evidence, their positions may be used in numerous situations. Some of these would be as an aid to proportion lost corners or to use as possible positioning of the corners "intersecting" these original lines to locate the "most probable" corner location, short of proportioning. Many of the original notes contain references to natural features or line trees, and after positive identification these may be utilized for the alignment of the section lines. In many older surveys, little evidence of these calls remains, and thus only by a careful and diligent search can it be said with certainty that the particular feature is the one called for.

10.20 PROPORTIONING: THE LAST RESORT

In football, "on fourth down and 50 yards to go," the quarterback asks the coach, "What do I do?" The answer probably will be, "When all else fails, you *punt*." This same question will be answered by many surveyors, attorneys, and courts, who have failed to apply a primary requirement in considering corners. The failure to ask the initial question "Is the corner lost or obliterated?" can pose serious problems. Before any discussion is conducted about any corner, the surveyor must address this important question.

Only when all of the evidence fails to prove a corner do you then proportion.

Any proportioning of lost corners should be considered as the last resort. It should be relied on only after an extensive search of the records, after conducting adequate measurements, after talking with people in local areas, and after much deliberation. In the case *U.S.* v. *John Citko*,[8] the federal judge in Wisconsin isolated the question to: Was the corner lost or obliterated? The United States held that the corner was lost, and thus it was proper to relocate it by single proportion. The defendants presented evidence based on witnesses' statements and the location of a rock mound, on the ground. Only when all of the evidence fails to prove a corner do you proportion. It has been found that many younger surveyors are quite innovative in locating or resurveying lost corners. Although in metes and bounds states there are no absolute guidelines, the GLO states have criteria that are strict yet tempered with flexibility. It is suggested that metes and bounds surveyors should understand the principles of locating lost GLO corners and apply the pertinent ones to their practices. This statement is particularly true now that many of the younger surveyors have ready access to electronic stations that permit easy measurements of angles and distances and make computations with little or no effort as well as GPS units. Relying on measurements as the primary methodology for ascertaining corners will soon dull the surveyor's ability to recover the evidence and evaluate this evidence that proves the existence or nonexistence of original corners.

Once again, it should be stressed that no proportioning should be applied until it is absolutely determined that a corner is absolutely lost. It must also be stressed that no proportioning between found corners can be predicated on corners located outside the boundaries of the township in a GLO state or the parent parcel of a metes and bounds state. All proportioning must be tied to two or more proven corners found within the original unit.

How lost is lost? The case of *U.S.* v. *Doyle* states that "for corners to be lost, they must be so completely lost that they cannot be replaced by reference to any existing data or other sources of information and before courses and distances can determine boundary, all means for ascertaining location of the lost monuments must first be exhausted."[9] Too many times, surveyors, and others, give up too quickly and set a new point by proportionment. This, at the least, may put a cloud on the resulting titles or, worse, cause resulting titles to be void, since the determination does not fit the basic requirement of the law.

10.21 RELOCATING LOST CORNERS

Principle 13. *Lost corners may be located by proportioning or by using a combination of evidence and measurements.*

As pointed out, lost corners are serious in that a proportioned point is presumed to be placed in its original position. One must apply a large degree of experience and common sense in using proportioning. The rules or guidelines do not permit proportioning if superior evidence is available. Assuming that positive evidence of an original section line location is superior to a proportioned point, there are no limitations on what can be used. Also assuming that a surveyor is able to positively identify an east–west fence line as being a section line by long-accepted reputation and testimony, that should be used. However, if a proportioned point would place the latitudinal point off the fence line, the surveyor would be well within acceptable survey practices to use the fence line as a latitudinal position and then rely on a proportioned longitudinal point.

> Courts prefer an obliterated corner based on "weak" evidence to a proportioned corner based on measurements alone.

Once it is determined that a corner is lost, the surveyor must decide whether it will be restored by federal or state rules. The first rules presented are an application of federal law. State laws that deviate from federal law are then identified. When a corner is positioned in accordance with the lost corner theory or proportioning, the surveyor must be convinced that no future surveyor or court will be able to locate the original corner in a different place. Proportionate measurement is always a rule of last resort and is always subject to question.

Principle 14. *The surveyor should strive to utilize all forms of evidence to prove a corner as an obliterated corner, rather than resorting to proportional measurements to set a lost corner.*

With the availability of surveying equipment and computer programs that make the measurements of angles and distances relatively easily, many surveyors are lulled into relying on the subsequent measurements and calculations to set new corners that may conflict with possession lines and may cause boundary disputes. The key between a lost corner and an obliterated corner may just be an interpretation of the recovered evidence and the reliance and weight that the retracing surveyor places on it, as well as the tenacity of the retracing surveyor.

It is difficult to argue with a measured line or a computed distance, yet it is very easy to disagree with the interpretation of the evidence relied on to position a corner. For this reason, the surveyor should be experienced in the interpretation of evidence., The surveyor must accept the realization that he or she can be firmly convinced about the authenticity of a corner, based on the evidence, but may find himself or herself having to take a defensive position with the "modern surveyor" who is armed with the latest in surveying measuring equipment and computer technology. Few young surveyors realize that it does not take $10,000 worth of surveying equipment to prove an original corner—the corner may be proved with a $10.95 shovel, used at the correct location.

> Only when the recovered evidence fails to prove a corner position can a surveyor resort to proportioning.

10.22 PROPORTIONATE MEASURE OR PRORATION

Principle 15. *When the surveyor has exhausted all possibilities of locating corners of public land surveys from available evidence, only then, as a last resort, should the surveyor rely on measurements alone.*

The retracing surveyor must strive to recover sufficient evidence on which to predicate a decision about the authenticity of the corner. If, after serious consideration and contemplation, the surveyor is still convinced that the evidence is insufficient to determine the validity of the corner, the surveyor must resort to measurements alone.

Principle 15A. *When proportionate measurements are used, the new values given to the several parts, as determined by the measurement, should bear the same relation to the record lengths as the new measurement of the whole bears to that record.*

Existing original corners (excepting closing corners) cannot be disturbed; consequently, discrepancies between the new measurements and those of the record will not in any manner affect the measurement beyond the corners identified, but the difference will be distributed proportionately within the several intervals along the lines between the corners.[10] In a subdivision where there is a difference in measurement between the original surveyor and the recent surveyor, it must be concluded that the difference occurred in all parts of the line unless the contrary can be proved beyond any reasonable doubt. In government surveys, many corners were set on a single line, whereas others were set checkerboard fashion with cross-ties. Single proportionate measure is applied to measurements along a single line, whereas double proportionate measure is applied to checkerboard surveys.

10.23 SINGLE PROPORTIONATE MEASUREMENT

Principle 15B. *A single proportionate measurement should be made between two proven corners, and may locate one or more points on a line.*

The classic example of single proportionate measure is the restoration of a lost quarter-section corner, as shown in Figure 10.2. Section corners *A* and *B* are known; the record distance from *A* to the lost quarter corner *Q* was 41 chains; the record distance from *Q* to *B* was 40 chains; the present measure from *A* to *B* is 81.50 chains.

Quarter corner *Q* is restored in the exact proportion of the original record; that is, (41 4 81) 3 81.50, or 41.26, chains is the distance from *A* to *Q* and (40 6 81) 3 81.50, or 40.24, chains is the distance from *Q* to *B*.

United States Code, Title 43, Section 752, states in part: "and the length of such lines (section lines) as returned shall be held and considered as the true length thereof." The original reported length is presumed to be the correct length, and any discrepancy in measurement that exists is presumed due to the recent surveyor. In resurvey, it is rarely found that the new measurement agrees with the old; hence, the new survey must be adjusted so that it will be in proportionate agreement with the

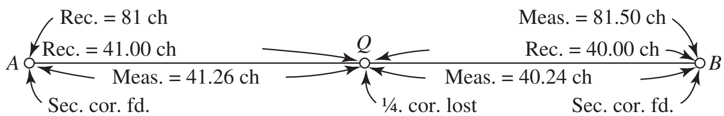


Figure 10.2 Single proportionate measure.

old survey. In other words, the new chain must be adjusted in length so that it agrees with the original measure. Actually, this is done by mathematics, as illustrated in the preceding example.

By law and by court ruling, original monuments, excepting closing corners, are fixed in position and cannot be moved. Proportionate measure cannot extend beyond any fixed point or fixed monument and is applicable only to lost points between original monuments.

Not all lines surveyed were straight lines. Corners on a standard parallel or east–west township lines were originally set on a curved line, and any restored corner that is located in the footsteps of the original surveyor must be on the same curved line.

10.24 DOUBLE PROPORTIONATE MEASUREMENT

Principle 15C. *Double proportionate measurement usually requires four original corners, and the location of the proportioned corners is positioned at the proportionate distances of lines run from the proportioned points. Primary weight is given to distance, not to angle or bearing.*

To restore a lost corner of four townships by double proportionate measurement, a retracement will first be made between the nearest known corners on the double proportionate measure meridional line, north and south of the missing corner, and upon that line a temporary stake will be placed at the proper proportionate distance; this will determine the latitude of the lost corner. Next, the nearest corners on the latitudinal line will be connected, and a second point will be marked for the proportionate measurement east and west; this point will determine the position of the lost corner in departure (or longitude). Then, through the first temporary stake run a line east or west, and through the second temporary stake a line north or south, as relative situations may determine; the intersection of these two lines will fix the position for the corner restored.[11]

During restoration of a corner by either single or double proportionate measurements, bearings or angles are of minor importance. Greater weight is placed on distance measurements, the reverse of what is usually true for federal metes and bounds surveys. In some states, the concept of distance controlling over direction may be in direct conflict with either state court holding or statute. On occasions, the instructions for double proportionate measurements have been misinterpreted. To obtain the intersection point, the instructions direct that the lines must be in cardinal directions, not at right angles to the lines. In most instances, the difference is slight, but occasionally it becomes significant.

Double proportionate measurement is illustrated by Figure 10.3, which shows the conditions in a hypothetical problem. To restore the lost section corner common to Sections 25, 26, 35, and 36 as shown in Figure 10.3, the surveyor located the southeast *(A)* and northwest corner *(B)* of Section 35, and the west quarter corner *(C)* and the southeast corner *(D)* of Section 25. A straight line from *A* to *C* is run and found to measure 120.60 chains, whereas the original recorded government measure was

40 chains for each half-mile, or a total of 120.00 chains. The surplus of 0.60 chain is divided into three equal parts of 0.20 chain each. A temporary point is established at *E*, exactly one-third of the way between *C* and *A* or 40.20 chains south of *C* on the line *AC*. On the east and west straight line between *B* and *D* is established a temporary point at *F*, the position of which is computed as follows:

$$\text{distance } BE = \frac{161 \times 80}{80 + 80.2} = 80.40 \text{ chains}$$

$$\text{distance } FD = \frac{161 \times 80.2}{80 + 80.2} = 80.60 \text{ chains}$$

From the proportionate point *E* a line is run due north (astronomical) and from the point *F* a line is run due west; their point of intersection determines the restored position for the lost section corner.

This is probably the main reason that surveyors who are trained in GLO principles will usually give primary weight to distances and secondary weight to angles or bearings.

10.25 RESTORATION OF LOST STANDARD CORNERS ON STANDARD PARALLELS, CORRECTION LINES, AND BASELINES

Principle 15D. *Lost standard corners will be restored to their original positions on a baseline, standard parallel, or correction line, by single proportionate measurement on the true line connecting the two nearest identified original standard corners. Proper adjustment should be made to secure the correct latitudinal curve. Closing corners are not to be used for either direction or measurement.*

The term *standard corners*, as used here, will be understood to mean all corners that were established on the standard parallel during the original survey of that line, including but not limited to, standard township, section, quarter-section, and meander corners. Closing corners or other corners purported to be established on a standard parallel after the original survey of that line will not control the initial restoration of lost standard corners.

In cases in which a closing corner was set on the standard parallel at the calculated position of the closing corner and at the time of the survey of the standard parallel, such a corner controls the direction of the standard parallel. For standard parallels, the standard corners set varied from time to time; that is, sometimes they applied to sections north of the line, at other times to sections south of the line.

Since corners originally set on standard parallels were set on a curved line, the lost corners must be restored on the original curved line connecting the nearest identified standard corner on each side. Lost corners on curved lines are restored by single proportionate measure on the curved line by first setting a temporary proportional corner on a straight line, then measuring over the proper correction for Earth's curvature.

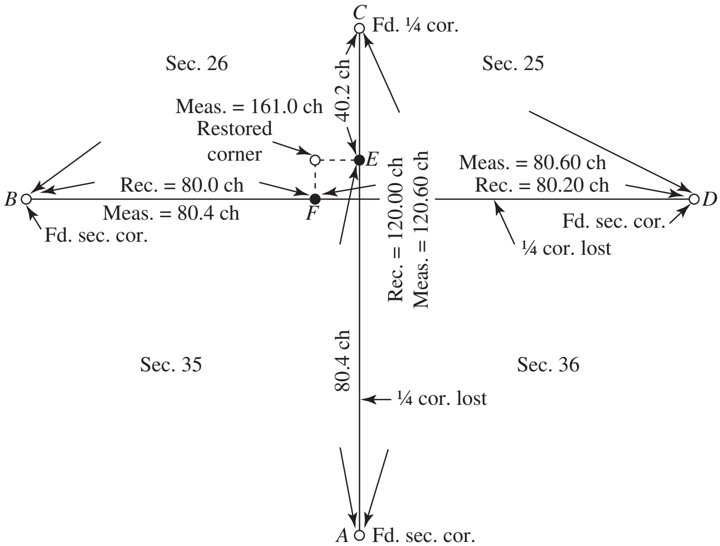


Figure 10.3 Double proportionate measure.

10.26 RESTORATION OF LOST TOWNSHIP CORNERS ON PRINCIPAL MERIDIANS AND GUIDE MERIDIANS

Principle 15E. *When the principal meridian or guide meridian was established by alignment in one direction only, lost township corners on such lines should be restored by single proportionate measurement. Where guide meridians were established as part of the scheme of township boundaries, all surveyed in the same system under a single contract, the township corners located thereon should be relocated by double proportionate measurements.*

When guide meridians were run as independent lines before the subdivision of township lines, cross-ties did not exist and restoration work should proceed on the basis of single proportionate means. However, under most contracts, guide meridians were established as part of the general scheme for the township boundaries with cross-ties to other township lines. When this occurred, double proportion should be used to restore lost township corners.

10.27 RESTORATION OF LOST TOWNSHIP AND SECTION CORNERS ORIGINALLY ESTABLISHED WITH CROSS-TIES IN FOUR DIRECTIONS

Principle 15F. *When lost regular township corners common to four sections and lost regular section corners between township lines were originally established with ties from four directions, the lost corners will be reestablished by double proportionate measurements.*

This rule is applicable to township corners not set on a baseline, principal meridian, and, occasionally, guide meridians. Most guide meridians were originally established as an integral part of the establishment of the township lines, with cross-ties in four directions; hence, lost township corners on guide meridians are usually restored by double proportionate means. The rule is also applicable to most lost section corners common to four sections and located between township lines. Where lost section corners between township lines were established by alignment in one direction only, such as corners along a sectional correction line or sectional guide meridian, single proportionate measure will be used.

Although the original east–west township lines were run on a curve, it is not necessary to rerun the lines on a curve where double proportion is used, as the final results will be identical whether curved or straight lines are employed. That is because the differences between the two are insignificant. Proportionate methods cannot extend beyond an identified original corner regardless of whether it is a quarter corner, section corner, meander corner, or a rarely set sixteenth corner. Once a corner is found undisturbed, its position is fixed and cannot be altered by proportionate means.

10.28 RESTORATION OF LOST CORNERS ALONG TOWNSHIP LINES

Principle 15G. *All lost section and quarter-section corners on township boundary lines will be restored by single proportionate measurement between the nearest identified corners on opposite sides of the missing corner, north and south on a meridional line, or east and west on a latitudinal line, after the township corners have been identified or relocated. An exception to this principle will be noted in the case of any exterior whose record shows deflections in alignment between the township corners.*

The strength of this principle lies in the fact that township lines were established before the subdivision of sections within a township. The corners being originally set on a line without cross-ties should be reestablished on the same line without cross-ties. The east–west township lines were curved lines, and the relocated lost corners must also be placed on a curved parallel of latitude. Single proportion cannot extend beyond a township corner that must be reestablished by double proportionate means.

10.29 RESTORATION OF LOST TOWNSHIP AND SECTION CORNERS WHERE THE LINE WAS NOT ESTABLISHED IN ONE DIRECTION

Principle 15H. *When a line has not been established in one direction from the missing township corner, the record distance will be used to the nearest corner identified in the opposite direction.*

This principle is applicable where double proportion normally would be used but cannot be applied because a line was not established in one direction. In Figure 10.4, the northwest section corner, the north quarter corner, the southeast section corner

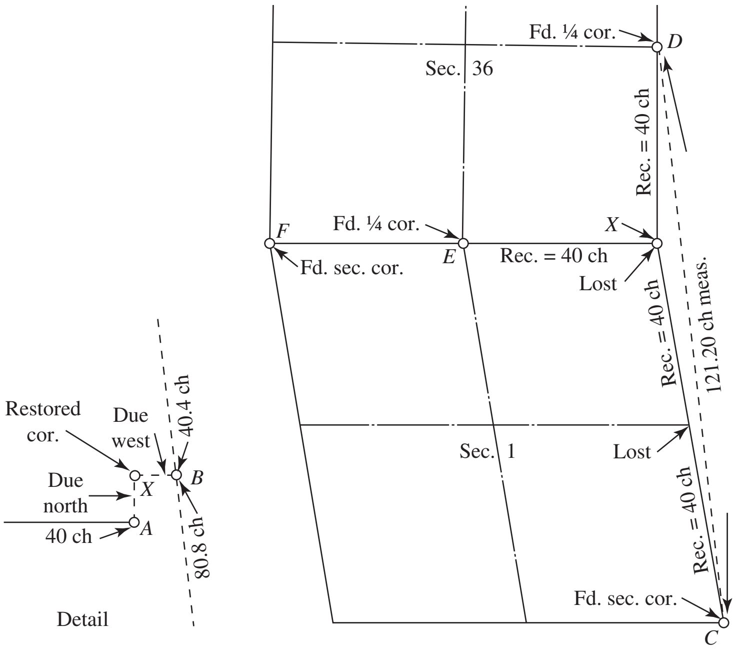


Figure 10.4 Restoration of a lost township corner originally established in three directions.

of Section 1, and the east quarter corner of Section 36 are all known. The townships to the east were not surveyed. Corner *X*, the lost township corner, is to be relocated. Run a line exactly 40 chains the record distance, due east from point *E*, and set a temporary point at *A*. Because due east means due east in the direction called due east by the original surveyor, the prolongation of line *F–E* would be considered due east. Next, set a point proportionate measure at *B* on a straight line from *D* to *C*, or 80.80 chains from *C* proportionate measure. Run a line from *B* due west and run a line from *A* due north. Their point of intersection at *X* is the lost corner.

10.30 RESTORATION OF LOST CORNERS WHERE THE INTERSECTING LINES HAVE BEEN ESTABLISHED IN ONLY TWO DIRECTIONS

Principle 15I. *Where the intersecting lines have been established in only two directions, the record distances to the nearest identified corners on those two lines will control the position of the temporary points; then from the latter, cardinal offsets will be made to fix the desired point of intersection.*

This principle is applicable in Figure 10.5, where the north and east quarter corners are known and the northeast section corner (also township corner) is to be set. The other townships were never surveyed. Run a line 40 chains due east from *C*, and set a temporary point A. Run a line 40.0 chains (record distance) due north from D, and set a temporary point at B. Run a line due west from B and a line due south from A. The point of intersection X is the lost corner.

10.31 RESTORATION OF QUARTER-SECTION CORNERS IN REGULAR SECTIONS

Principle 15J. *All lost quarter-section corners on section boundaries within the township will be restored by first establishing or proving the section corners on each side; then the quarter corners will be relocated by single proportionate measurement on line. An exception occurs where the original lines had angular deflection.*

The importance of one line over another is recognized by this rule, in that section corners must be relocated before the restoration of a lost quarter corner. Lost regular quarter corners, whether located on standard parallels, sectional correction lines, or any other line, are restored by single proportionate measure between the section corners located on each side of the missing quarter corner. Where the original line run was a curved parallel of latitude, the relocated quarter corner must also be placed on the original curved line. If the quarter corner was originally placed on a line that showed angular deflections, the restored corner must be relocated with the angular deflection.

10.32 RESTORATION OF QUARTER-SECTION CORNERS WHERE ONLY PART OF A SECTION WAS SURVEYED ORIGINALLY

Principle 15K. *Where a line has been terminated with measurement in one direction only, a lost corner will be restored by record bearing and distance, counting from the nearest regular corner, the latter having been duly identified or restored.*

In applying this principle, some surveyors may tend to give it more credence than it deserves. This should be considered as the last resort, yet it places a technical responsibility on the retracing surveyor. Because the lost corner is dependent on the bearing and the distance, there is no leeway in applying the distance. Under this rule, the distance controls, but there is latitude in the bearing on which the line is to be measured.

There are other questions that should be addressed: Is the bearing of the line supposed to be the cardinal or true bearing that is indicated in the notes or the plat? Or can it be a mean of the two lines that lie on either side of the line? If a chaining index is determined, can the surveyor apply this chaining index to the record bearing that is to be placed on the ground? Unfortunately, these questions cannot be answered, because there are no definite answers to them. The ultimate decision will be made by the retracing surveyor, who must ultimately defend any methods used.

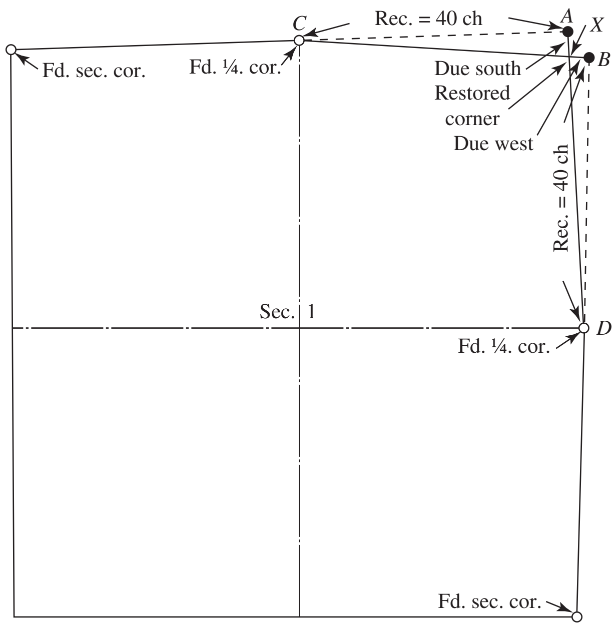


Figure 10.5 Restoration of a township corner originally established in two directions.

10.33 RESTORATION OF A CLOSING SECTION CORNER ON A STANDARD PARALLEL

Principle 15L. *A lost closing corner on a standard parallel will be reestablished on the true line that was closed upon, and at the proper proportional interval between the nearest regular corners to the right and left. The only corners that will control the direction of the line being closed upon are (1) standard township, standard section, and standard quarter corners; (2) meander corners terminating the survey of the standard parallel; and (3) closing corners in those cases where they were established originally by measurement along the standard line as points from which to start a survey.*

Figure 10.6 shows a standard parallel and Sections 2 and 3, whose lines closed upon the parallel when surveyed from the south toward the north. The corner common to Sections 2 and 3 is to be restored by proportionate means. Because the lost corner was originally set on the standard parallel, it must be reset upon the line at a proportionate distance from the next regular corner to the right and left. Closing corners were not set at the time at which the standard parallels were run and cannot be considered as determining the direction of the parallel; only those set as part of the standard parallel when it was run originally may be used. In the preceding example, the record distance from the closing corner to the nearest regular corner to the east was 30 chains and 10 chains to the quarter corner to the west. Recent measure revealed a 1-chain surplus between the found regular corners, which is divided into 0.75 chain and 0.25 chain by proportion so that the new distance to the regular section corner to the east becomes 30.75 chains.

10.34 RESTORATION OF A LOST NORTH QUARTER CORNER IN A CLOSING SECTION

Principle 15M. *When the north quarter corner on a closing section was originally set and lost, the lost corner will be reestablished on the closing line at a point at the proper proportionate interval between the nearest found or relocated corners to the right and left.*

The north quarter corner of a closing section on a standard parallel was not normally set.

Principle 15N. *The method to be followed in the subdivision of a section into quarter sections is to run straight lines from the established quarter-section corners*

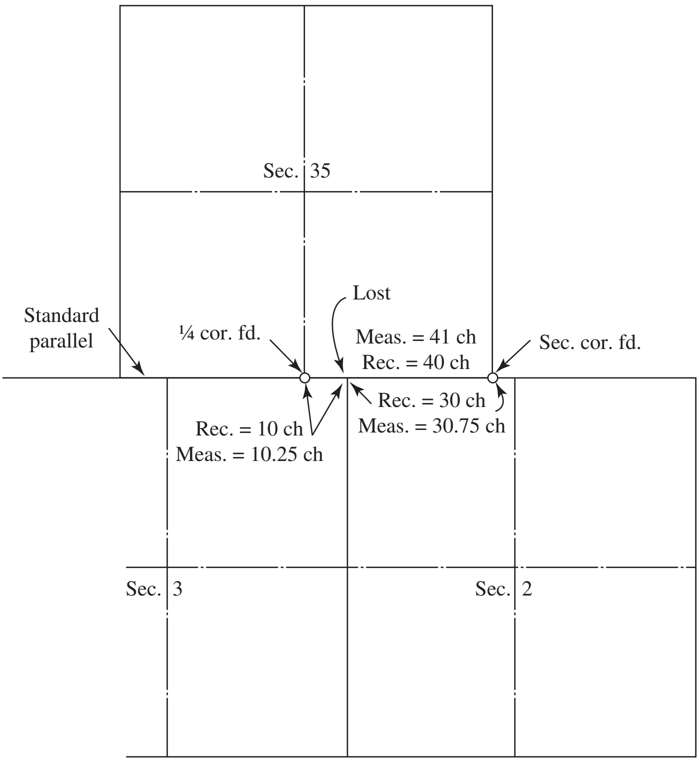


Figure 10.6 Restoration of a lost section corner on a correction line.

to the opposite quarter-section corners; the point of intersection of the lines thus run will be the corner common to the several quarter sections, or the legal center of the section.

This principle probably is the one discussed most. Although this principle is established in law, the Land Act of 1805, practicing surveyors deviate from it considerably. Over the years, numerous short cuts were taken to locate the *legal center of the section*, yet none were sanctioned by the law. In retracing past surveys that were conducted improperly, if the modern surveyor conducts his or her work properly, deviations would be found in possession lines and surveyed lines. However, the modern surveyor is obligated to conduct the modern work according to the law and principles. There can be no deviation. If, in conducting a modern survey in accordance with proper rules and principles, the surveyor determines possession lines, and surveyed lines disagree on the center of the section, the only obligation the retracing surveyor has is to report these variances to the client, not to ascertain property rights to the mislocated fences.

Principle 15O. *Preliminary to the subdivision of quarter sections, the quarter-quarter or sixteenth-section corners will be established at points midway between the section and quarter-section corners, and between the quarter-section corners and the center of the section.*

Principle 15P. *The law provides that where opposite corresponding quarter-section corners have not been or cannot be fixed, the subdivision of section lines shall be ascertained by running lines from the established corners north, south, east, or west, as the case may be, to the watercourse, reservation line, or other boundary of such fractional section, as represented on the official plat.*

10.35 RESTORATION OF LOST NONRIPARIAN MEANDER CORNERS

The original surveyors also set a special type of corner that does not identify specific sections. In later surveys, these *meander corners* were set at points where meander lines were run to identify the exterior of topographic features, usually water bodies, cross-section lines, or township or range lines. In most instances, no bearing tees were set, but the point was monumented. Meander corners can be used for line direction as well as proportioning.

The wording of the original surveyors in the field notes becomes critical when it comes to whether a line is a meander line or not. The application of survey principles to meander lines and non-meander lines is totally different. Read the field notes, patents, and all related documents is the best advice that can be given.

The following principle applies to meander corners.

Principle 16. *Lost meander corners, originally established on a line projected across the meanderable body of water and marked on both sides, will be relocated by single proportionate measurement after the section or quarter-section corners on the opposite sides of the missing meander corner have been duly identified or* *relocated. Where a line was terminated with measurement in one direction only, a lost corner will be restored by record bearing and distance, counting from the nearest regular corner, the latter having been duly identified or restored.*

The actual shoreline of a body of water is considered the correct terminus of a line; the meander corner controls the direction of the line. The restoration of a lost meander corner would be required infrequently.

10.36 RESTORATION OF RIPARIAN MEANDER LINES

Principle 17. *A meander line platted as the boundary of a riparian owner is seldom the true boundary; the body of water represents the true boundary. Lands acquired by the Swamplands Act and fraudulent surveys go to the meander line.*

The purpose of meandering any body of water was to obtain information that would aid plotting the body of water on maps. The shoreline itself is the best evidence of the true location and will govern.

When a survey was fraudulent or grossly inaccurate in that it purported to bound tracts of public lands on a body of water when in fact no such body of water existed at or near the meander line, the false meander line, not an imaginary line to fill out the fraction of the normal subdivision, marks the limits of the grant of a lot abutting thereon, and, upon discovery of the mistake, the government may survey and dispose of the omitted area as a part of the public domain.[12] Swamplands conveyed to the state under the Swamplands Act are generally limited by the meander line, as no definite riparian waterline exists.

10.37 RESTORATION OF NONRIPARIAN MEANDER LINES

Principle 18. *After locating the positions of the meander corners on section lines, the record meander courses and distances are run, setting temporary angle points; the closing error from the last meander course to the section line meander corner is noted. The error of closure is balanced out by the compass rule.*

This principle is attained in the field by moving each temporary angle point or location on a bearing of the closing error a distance in proportion to the length of the closing error as the sum of the courses to the angle point is to the sum of all the courses. Although this method of locating meander points is valid in GLO states, the principle can be applied to a metes and bounds description when a surveyor determines several corners in a description as lost. Where the angle points in the meander lines are considered corner points, either approach applies.

The method used is illustrated in Figure 10.7. The resurvey, beginning on the west side of the section at the found or reestablished meander corner, is run eastward on the bearing and distances of the original notes. After setting temporary stakes at each angle point, the closing bearing and distance from the last temporary stake to the true meander corner is noted, or, as illustrated, N 20° E, a distance of 30 feet. From

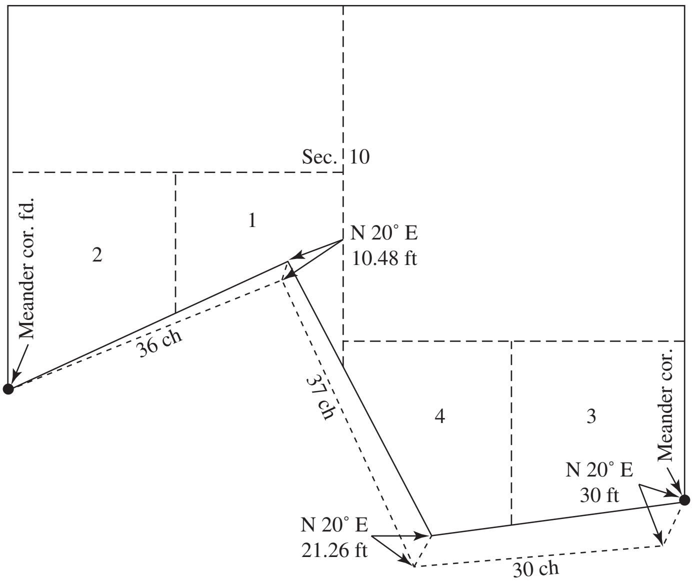


Figure 10.7 Restoration of a lost nonriparian meander line.

each temporary stake, the true corner is set on the bearing of the closing line a proportionate distance. The first temporary point would be moved 30 times 36/103, or 10.48 feet, as illustrated. Exactly the same results are obtained by using the compass rule of adjustment.

10.38 RESTORATION OF IRREGULAR EXTERIORS

Principle 19. *Where boundaries that would normally be straight lines were originally established in an irregular manner, the resurvey (retracement) will follow the irregular procedure.*

Where a township line was partially surveyed and later completed by surveying from another direction, an angle point at the junction of the two lines would be perpetuated in resurvey methods. Recent retracements of township lines for the purpose of resurveying a township frequently uncover deflections in alignment along the township lines. New corners set upon such lines have an original-record angular alignment and must be resurveyed taking the angular deflections into account.

10.39 LOST CORNER RESTORATION METHODS

A summary of the usual methods by which surveyors should restore lost corners within a standard township surveyed in accordance with *Manuals* after 1855 is

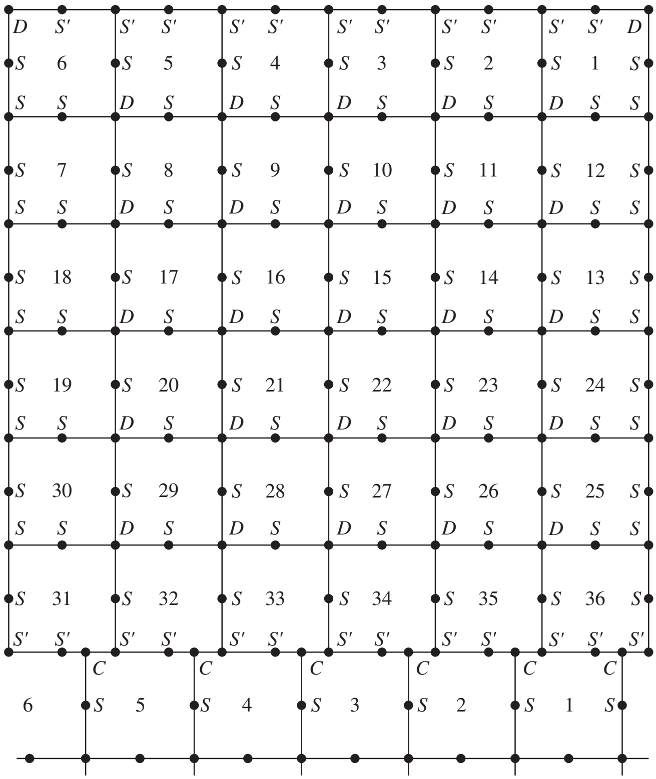


Figure 10.8 Chart showing the normal methods of restoring lost corners. *S*, single proportion; S' single proportion on a curved parallel; D, double proportion; C, closing corner.

outlined in Figure 10.8. This diagram may not be applicable where irregular original procedures were used or where early instructions deviated from those approved in published *Manuals*. For example, corners on a sectional correction line acquire the status of a township line; lost corners are restored by single proportionate measurements. The letter *D* indicates that double proportionate measurement is used. The letter *S* indicates that single proportionate measurement is used. The letter *S*9 indicates that single proportionate measurement on a curved parallel is used. The letter *C* indicates a closing corner that is to be adjusted to the line closed on. The most important rule to remember is: Proportionate measurement is a rule of last resort; don't use it merely because it requires less work.

10.40 RESURVEY INSTRUCTIONS ISSUED IN 1879 AND 1883

On November 1, 1879, the General Land Office of the Department of the Interior issued a circular signed by J. M. Armstrong, Acting Commissioner, which stated:

"There being no special law in regard to the reestablishment of lost corners, the rule (as given) is to be considered merely as an expression of the opinion of this office." With regard to lost section corners, the circular said: "Missing section corners in the interior of townships should be reestablished at proportionate distances between the nearest existing original corners NORTH and SOUTH of the missing corner." Double proportioning was not used!

In the pamphlet entitled *The Restoration of Lost and Obliterated Corners*, published in 1883, the instructions were to use double proportioning to relocate lost interior section corners. We can only wonder how many lost section corners were set by single proportionate measurements during the period between 1879 and 1883.

10.41 HALF-MILE POSTS IN FLORIDA AND ALABAMA

Two states, Florida and Alabama, were treated to the unique situation of half-mile posts set or established at 40.00 chains along east and west or north and south lines, as described in Section 6.14. Part of the time, the half-mile posts happen to fall where the quarter corner should be. Sometimes the quarter corner was set; at other times it was not. In all instances, the half-mile post was placed on the random line, and if it happened to fall where the quarter corner should be, it became the quarter corner.

In a retracement, the half-mile post, when found, is always used as a control point but not always as the quarter corner.

Some notes may indicate that the original surveyor returned and corrected the half-mile post to the true quarter corner position without new ties to bearing trees. Thus, if the quarter corner post is lost, the bearing trees relocate the half-mile post, not the quarter corner. The half-mile post, being a fixed point on the original survey, can be used as a proportioning point to restore adjoining lost section corners. Proportionate measurements can never extend beyond original points identified. If an adjoining lost section corner is relocated by double proportionate measurement, the quarter corner is located by single proportionate measurements utilizing the half-mile post.

The evidence of the half-mile post should never be destroyed by the retracement surveyor, nor should the surveyor assume, without checking the field notes, that the half-mile post is the quarter corner—often it is not.

SUBDIVISION OF SECTIONS

10.42 GENERAL COMMENTS

The present federal rules for subdividing sections are clear; the date of their first application is questionable. Certainly, these rules could not have been formed prior to the time the Land Office was empowered to sell land in portions of a section. In

1805 the law provided for land to be sold by quarter sections, in 1820 in half-quarter sections (80 acres), and in 1832 in quarter-quarter sections (40 acres). Prior to 1820 the practice of dividing sections by protraction on township plats did not exist. In 1879, J. M. Armstrong of the Land Office issued a pamphlet on how to restore lost corners and subdivide sections. His method of dividing a section into quarter-quarter sections agrees with what is accepted today. In older areas, where the government did not protract portions of sections and where land was later sold by aliquot parts, the surveyor should consult state court findings to obtain subdivision guidance.

Prior to the issuance of the pamphlet *Restoration of Lost and Obliterated Corners and Subdivision of Sections*, and even afterward, many surveys conducted by private parties or self-proclaimed surveyors were accomplished by measuring lines north, south, east, or west in increments of 1320 feet (20 chains). Such procedure is, of course, entirely wrong; no portion of a section can be laid out correctly without first resurveying the entire section.

10.43 SUBDIVISION BY PROTRACTION

Sections of land, being the minimum area normally surveyed, are too big to completely serve the purpose of disposal of public lands. Where an entryman may patent a quarter of a quarter of a section, the section must be subdivided further to allow proper delineation of the land boundaries; the process is known as *subdivision by protraction*. Upon the original township plats are shown dashed lines that indicate the intent of the government in disposing of parcels less than a section in size. Together with the federal land statutes or regulations of the Bureau of Land Management and plats subdivided by protraction, the surveyor can determine the proper procedure for the survey of portions of sections.

Regular sections of land as shown on the official plats have straight dashed lines connecting opposite quarter corners to indicate that quarter sections are to be subdivided by running lines connecting opposite quarter corners. Further dashed lines to indicate smaller divisions of land are omitted except where irregular conditions exist, as shown in Figure 10.9. All quarter-quarter section lines not shown on the plat are assumed to be of regular size, as required by statutes. Irregular-sized parcels and quarter-quarter parcels containing irregular measurements are given lot numbers on the plat to distinguish them from regular parcels.

10.44 ESTABLISHING THE NORTH QUARTER CORNER OF CLOSING SECTIONS ON A STANDARD PARALLEL AND OTHER QUARTER CORNERS NOT ORIGINALLY SET

Principle 20. *Where an original quarter-corner was not originally set, as in the case of sections closing on a correction line or sections closing on a range line with double or triple corners or sections closing on a township line with double or triple corners, place the missing corner on the correction line (or range or township line)*

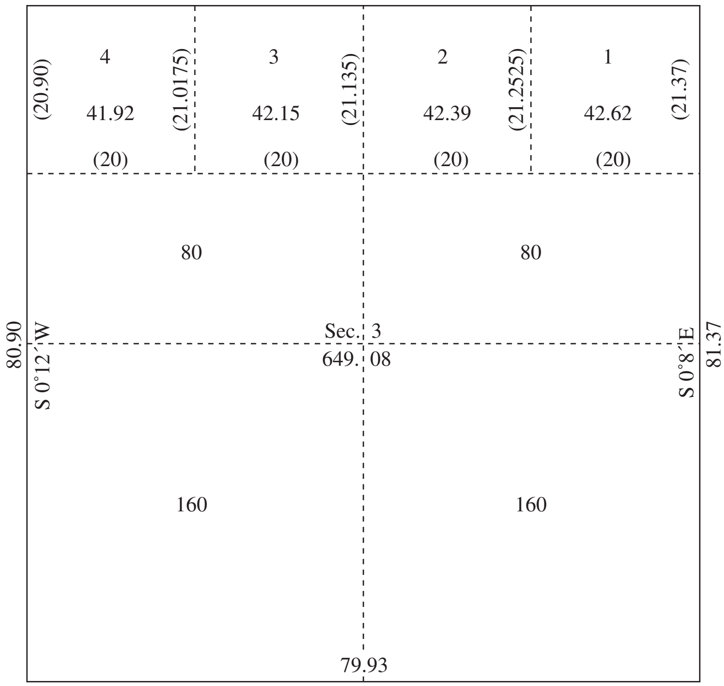


Figure 10.9 Dashed lines indicating smaller divisions are used only when irregular conditions exist.

at a point between the found or relocated closing section corners a distance that is proportional to the measurements used for the acreage calculations on the original plat. The missing corner is usually set midway between closing section corners except in Section 6, where it is usually 40 chains proportional measure from the northeast or southwest closing section corner.

By an older principle, when a township was being subdivided, the last section lines, run west to intersect the west line of the township, did not have to meet the township line at the same point as the formerly set section corner. Double corners existed: the original section corner for sections west of the township line and the newly set closing section corner for sections east of the township line. By such a rule, the quarter corners for the sections east of the township line—6, 7, 18, 19, 30, and 31—were never set. The quarter-section corner found is applicable only to the sections west of the township line. In the event of triple corners, the quarter corner on the township line was not applicable to either section other than as a line point.

The north quarter corner of Sections 1 to 6, inclusive, was seldom set where the north line of the township was a correction line. Prior to establishment of the missing quarter corner, the closing section corners found must be tested for alignment. If found off the line closed upon, they must be moved up to the line closed upon. From the acreage figures on the original plat, the distances used for computing the original acreage can be reconstructed. For the purpose of illustration, Section 3 shown in Figure 10.9 gives the original measurements of a section whose north quarter corner was never set. Figures in parentheses are those computed by the rules of protraction. Since 1 acre is 10 square chains, the acreage of Lot 1 is $20 \times (21.37 + 21.2525) \div 2 = 426.2 \div 10 = 42.62$ acres. Where the original width of Lot 1 was 20 chains, the acreage was determined by the sum of the lengths of the east and west lines. Conversely, where the sum of the east and west sides equals the acreage, 20 chains was the assumed original width.

Because each of the lots in Figure 10.9 can be proved by computation to be 20 chains wide, the missing quarter corner would be set midway between the found or relocated closing section corners to the east and west.

10.45 ESTABLISHMENT OF CENTERLINES AND CENTER QUARTER CORNERS

Principle 21. *The method to be followed in the subdivision of a section into quarter sections is to run straight lines from the established quarter-section corners to the opposite quarter-section corners; the point of intersection of the lines thus run will be the corner common to the several quarter sections, or the legal center of the section.*

Boundary lines that have not actually been run and marked shall be ascertained by running straight lines from the established corners to the opposite corresponding corners.[13] Federal laws are very specific and unambiguous; the legal center of a regularly created section is at the exact point of intersection of the two lines running from opposite quarter-section corners. This rule does not require a section to be divided into two halves of equal area. Although a monument found near the legal center may be controlling for property rights because of occupancy rights, it is never the legal center of the section.

Of course, if the *original surveyor* used an irregular procedure and set a monument at the center of a section, the monument, wherever set, will be the center of the section. Also, if half of a section were created by one original survey and the remaining half by a dependent survey, the rule may not apply because of the irregular original procedure. From 1849 to 1851, Justin Butterfield, Commissioner of the General Land Office without authorization, issued special instructions that directed that the center-quarter corner would be located at the midpoint of the line connecting the east quarter-section corner and the west quarter-section corner. If it can be shown that during those years these instructions were followed, the corner so set may be the legal center of the section.

When in a retracement the surveyor finds a use and occupation monument inconsistent with the legal center of the section, a dilemma arises: Should it be cast aside? Certainly, the monument can never be the legal center of the section, but property rights may have been lost or gained by unwritten means, or conveyances may have been completed to the monument. When an improperly placed monument is found, the surveyor must investigate the various possible legal ramifications that present themselves.

For some unexplainable reason, the center of the section seems to enter into the discussion in conversations with surveyors at annual meetings, parties, and other occasions. At times, the discussions become quite opinionated. The young surveyor must make a distinction between "Where is the center of the section?" and "What is the center of the section?" In nearly all sections that have been surveyed, a retracing surveyor, after locating the intersection of the lines as dictated by the law, will find a fence corner or monuments at positions that disagree with the intersecting lines or at a location other than the intersection. The distance may range from a few inches to chains. There are surveyors who are willing to accept these monuments as being the center of the section. This problem has plagued surveyors and the courts for over 200 years. The problem still has not been solved.

In a recent Oregon decision, *Dykes* v. *Arnold*, 129 P3d 257 (2006), the Supreme Court totally disregarded the law and created a "new center of the section some 75 feet away from the true intersection of the two lines by declaring that the "new center of the *section*" had been created by a doctrine of repose. The question that persists is why did they exceed their authority to take this route when they could have declared the fence corner a "property corner"? Are future surveyors who will subdivide the section into smaller parcels required to use this new center section to further subdivide?

These monuments may ultimately be considered as property corners, but there is no manner in which they may be considered as the center of the section. The law is specific and unyielding. On several occasions, some surveyors have attempted to justify the acceptance of these monuments by calling them by such names as "as-built corners" or "accepted center section corners." Their acceptance or rejection are not technical questions but legal questions.

10.46 ESTABLISHMENT OF QUARTER-QUARTER SECTION LINES AND CORNERS

Principle 22. *Preliminary to the subdivision of quarter sections, the quarter-quarter or sixteenth-section corners will be established at points midway between the section and quarter-section corners, and between the quarter-section corners and the center of the section, except on the last half-mile of the lines closing on township boundaries, where they should be placed at 20 chains, proportionate measurement, counting from the regular quarter-section corner. The quarter-quarter or sixteenth-section corners having been established as directed in these principles, the center-lines of the quarter sections will be run straight between opposite corresponding quarter-quarter or sixteenth-section corners on the quarter-section boundaries. The intersection of the lines thus run will determine the legal center of a quarter section.*

"In every case of the division of a quarter section the line for the division thereof shall run north and south . . . on the principles directed and prescribed by the section preceding."[14] The preceding section states: "And the boundary lines which have not been actually run and marked shall be ascertained by running straight lines from the established corners to the opposite corresponding corners."

This rule is applicable in most states, but there are several exceptions. In Louisiana and Mississippi, the original surveyors laid out some of the townships with the error of closure being divided equally between the last two closing half-miles (halfway between the two section corners). In several other states, particularly in Florida and Alabama, the error of closure was sometimes *placed* in the south half mile of the southern tier of sections. Of course, in situations like this, the usual rules of retracement will not apply.

10.47 FRACTIONAL SECTIONS CENTERLINE

Principle 23. *The law provides that where opposite corresponding quarter-section corners have not been or cannot be fixed, the subdivision-of-section lines should be ascertained by running lines from the established corners north, south, east, or west, as the case may be, to the watercourse, reservation line, or other boundary of such fractional section, as represented on the official plat. In this, the law presumes that the section lines are due north and south, or east and west lines, but this is not usually the case. Hence, to carry out the spirit of the law, in running the centerlines through fractional sections, it will be necessary to adopt mean courses, where the section lines are not due cardinals, or to run parallel to the east, south, west, or north boundary of the section, as conditions may require, where there is no opposite section line.*

In those portions of the fractional townships where no such opposite corresponding corners have been or can be fixed, the boundary lines should be ascertained by running from the established corners due north and south or east and west lines, as the case may be, to the watercourse, Indian boundary line, or other external boundary of such fractional township.[15]

This statute is interpreted by the federal courts, as noted. In a few states, the courts say that the east–west centerline of a section should be parallel with the south boundary of the section. The federal rule is preferred.

10.48 SENIOR RIGHT OF LINES

The principle of the precedence of one line over another of less original importance has long been recognized. Like property rights, surveys also have senior and junior rights or dignity. For example, if Township 1 is surveyed and accepted prior to the later survey and acceptance of Township 2 and it is found that Township 2 encroaches on Township 1, then as the senior survey, Township 1 will receive full measure. All closing corners, representing a junior survey, are adjusted to the line closed on.

This has led to the legal question "May a surveyor cross a township line to proportion lost corners?" The answer is a resounding *no*. (See Section 10.50.)

10.49 GROSS ERRORS AND ERRONEOUSLY OMITTED AREAS

When originally establishing sections of land by survey, gross errors did occasionally occur. This was particularly true of subdivisions adjoining rancho land grants, Indian treaty lines, and the meander of lakes or waterfronts. Where the federal government sold land not its own (overlaps or rancho land grants in particular), the error can be resolved by senior rights or by prolonged occupancy with color of title.

Areas omitted present a complex problem, as occupancy rights against the federal government cannot ripen into a fee title. Omissions of small areas (failure of a closing corner to close on the line called for) cause no problem, as it is a well-established rule that erroneously located closing corners may be adjusted to the original line closed on.

In the event of omission of islands that existed as of the date of the original survey, the U.S. government can claim the islands as unsubdivided public domain land. Many present-day surveys by the Bureau of Land Management are for the purpose of mapping omitted islands in rivers and lakes.

In several instances, the original surveyors included meanders of nonexistent lakes or exaggerated meanders of existing lakes (presumably, to get extra pay for mileage of lines run). Figure 10.10 shows the meanders of Moon Lake (T.12 N., R. 9 E., Fifth Principal Meridian, Arkansas) as originally reported. Investigation on the ground disclosed that the greater part of the tract was covered with oak, maple, cottonwood, hickory, sycamore, hackberry, cypress, and willow and that no lake could have existed as of the date of Arkansas's admission to the Union. Arkansas had acquired the surrounding fractional sections by the Swamplands Act and was claiming the Moon Lake area as based on riparian rights. In rejecting the riparian claim of the adjoiners, the court commented as follows:

> *First*. Where, in a survey of the public contain a body of water or lake is found to exist and is meandered, the result of such meander is to exclude the area from the survey and to cause it as thus separated to become subject to the riparian rights of the respective owners abutting on the meander line in accordance with the laws of the several States. *Hardin* v. *Jordan*, 140 U.S. 371; *Kean* v. *Calumet Canal Co*., 190 U.S. 452, 459; *Hardin* v. *Shedd*, 190 U.S. 508, 519.
>
> *Second*. But where upon the assumption of the existence of a body of water or lake a meander line is through fraud or error mistakenly run because there is no such body of water, riparian rights do not attach because in the nature of things the condition upon which they depend does not exist and upon the discovery of the mistake it is within the power of the Land Department of the United States to deal with the area which was excluded from the survey, to cause it to be surveyed and to lawfully dispose of it. *Niles* v. *Cedar Point Club*, 175 U.S. 300; *French-Glenn Live Stock Co*. v. *Springer*, 185 U.S. 47; *Security Land & Exploration Co*. v. *Burns*, 193 U.S. 167; *Chapman & Dewey Limited Co*. v. *St. Francis Levee District*, 232 U.S. 186.
>
> If in the making of a survey of public lands, an area is through fraud or mistake meandered as a body of water or lake where no such body of water exists, riparian rights do not accrue to the surrounding lands, and the Land Department, upon discovering

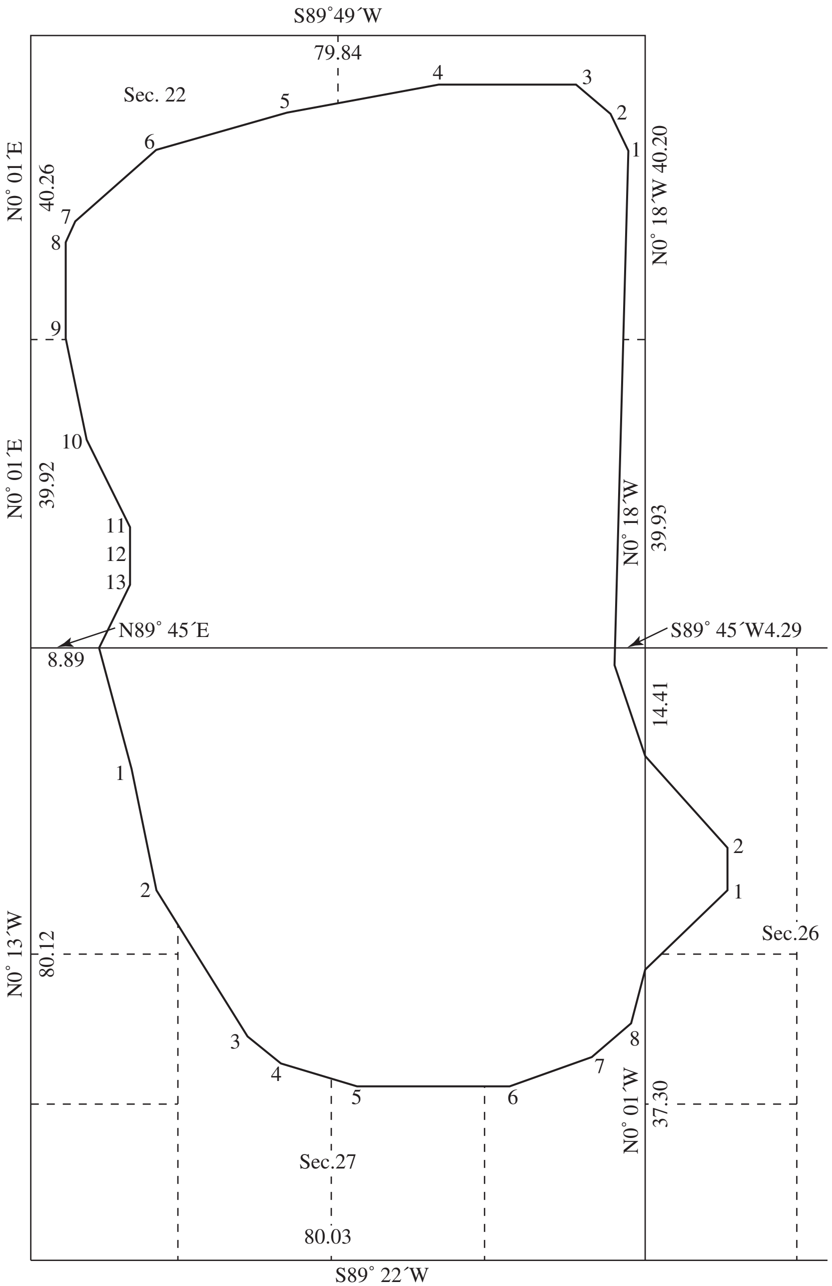


Figure 10.10 Moon Lake case.

the error, has power to deal with the meandered area, to cause it to be surveyed, and lawfully to dispose of it.

The fact that its administrative officers, before discovery of the error, have treated such a meandered tract as subjected to the riparian rights of abutting owners, under the State laws, and consequently as not subject to disposal under the laws of the United States, cannot stop the United States from asserting its title in a controversy with an abutting owner; and even as against such an owner, who acquired his property before the mistake was discovered and in reliance upon action and representations of Federal officers carrying assurance that such riparian rights existed, the United States may equitably correct the mistake and protect its title to the meandered land. The equities of the abutting owners, if any, in such circumstances are not recognizable judicially, but should be addressed to the legislative department of the Government.[16]

10.50 RELOCATING CORNERS FROM OTHER TOWNSHIPS OR FROM INTERIOR CORNERS

As has been pointed out, surveyors are innovators when it comes to using exotic, and possibly unauthorized or unaccepted, methods of locating or proving corners. In the basic scheme of authority in the subdivision of a section, sixteenth corners are located and surveyed from section and quarter-section corners. Thus, they are superior to the sixteenth corners, and the sixteenth corners should not be used to locate section corners or quarter-section corners. If, in the course of a survey, a question is presented as to whether an original corner is lost or obliterated, the measurement from a sixteenth corner may be used, along with all of the other evidence, as a means of attempting to prove the corner in question, but not to set it.

Applying the principle that each township is a separate and independent entity, one cannot cross that invisible township or range line to resurvey a corner in an adjacent township. The surveyor should rely on original corners to set or position lost corners; that is, the surveyor should not use a sixteenth corner to set a quarter corner. Except for those rare instances when an original surveyor set an interior corner within a section, the retracement surveyor should not go from within a section to set a corner along a section line. If, in the course of a survey, an original corner position is thought to be on the verge of a lost corner and additional evidence is needed to prove it to be obliterated, a measurement from a found monument within the section may be taken into consideration.

However, a found interior subdivision corner should not be accepted without being referenced to a found or proven original corner or without evaluating the field notes of the surveyor who subdivided the section.

10.51 PROCEDURES FOR CONDUCTING RETRACEMENTS

A surveyor should not undertake a retracement of a section or a township or a metes and bounds survey without first having an understanding of what is required. A retracement survey should be planned in advance to ensure that nothing is overlooked. The steps suggested for this procedure are as follows:

1. If possible, obtain necessary original field notes and township plats pertaining to the area being surveyed. If the state has more than one source, determine which source has the “official” field notes. For example, original California field notes were burned in the San Francisco fire and copies were made from the government copies. Thus, California now has copies of copies and the government copies are “official” documents. Included should be original surveys and any subsequent dependent or independent resurveys as well as official retracements. In several metes and bounds states, all official records were burned when Sherman made his infamous “march to the sea” during the Civil War in 1865. In many metes and bounds states, the original surveys were private in nature and the original notes were never turned over to the respective states.
2. Search known records for subsequent surveys conducted by private or public parties, such as road, railroad, and court-ordered surveys.
3. Contact old residents concerning ancient land boundaries. This may include taking affidavits to preserve testimony from elderly witnesses.
4. Examine recorded and known pertinent deeds of landowners along the lines to be surveyed and any recorded documents that may show easements or encumbrances.
5. Make a diligent search for all necessary corners and apply the rules of evidence to determine whether a corner is original, obliterated, or lost. If it is an obliterated corner, it should be refurbished with a durable monument with the surveyor’s identification numbers on it. If the corner is lost, reposition it according to the proper rules of survey.
6. Set new monuments for new positions and replace any monuments that have deteriorated.
7. Subdivide the section and set any required subdivision corners according to applicable rules of survey, whether federal or state.
8. Prepare and file a record of survey indicating the dignity of all points recovered or set in the retracement, and identify all new monuments set. Furnish the client with a written report of the methods used, monuments set, and basis of decisions made.

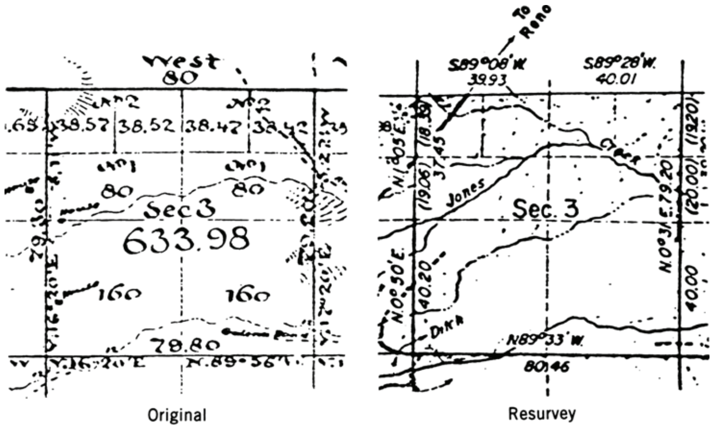
The responsibility of the retracement surveyor is to follow the evidence of the footsteps of the original surveyor as nearly as possible. However, these instructions should equate to “following the evidence of the footsteps.” This will require that the retracing surveyor go into the field armed with as much of the documentary evidence as possible and then try to determine or ascertain the location of the original footsteps through the recovered evidence. In light of modern technological advances, it is virtually impossible, surveywise, to follow the footsteps of the original surveyor. Using calculation to determine positions of the original lines, these original lines can be retraced. The retracing surveyor must always keep in mind the precision of the original surveys. The degree of positional tolerance from lines that are all measured to the full chain and the angles measured to the full degree are not absolute in position but they have "areas of certainty."

After recovering and evaluating the evidence available, the surveyor determines that there is no exact certainty as to the precise location of the exact corner point, but its corner can be located only within an area of certainty; for example, "The location falls within a circle of 9 feet in radius, based on the evidence." What some people would consider as a weakness is not, in that we can say: "The corner can be anyplace within this 6-foot circle but certainly is not some 300 feet over there."

10.52 INTERPRETATION OF ALIQUOT DESCRIPTIONS

Surveyors seldom survey or retrace more than a fractional portion of a section or sections. For the surveyor to understand what is required, the client usually furnishes the surveyor with a description, usually referring to an aliquot portion. Yet the description furnished is often far from a true aliquot description. Some of these descriptions may be as follows: "the east 80 acres of the NE ¼," "the west 1320 feet of the NW ¼," "the north 10 chains of the NE of the NW," "the NE ¼ lying north of Highway 6." In fact, one of the worst descriptions, which has no legal meaning, is "the NE ½ of the section"; there is no basis of validity for this description. These descriptions should not be considered as an aliquot description but are in all probability a modified (quasi) metes and bounds description. One could also encounter a description that calls for "lot 2, containing 80 acres." There could possibly be a conflict between "lot 2" and "80 acres."

In examining the original plat, the closing Section 3 would depict two lots, with Lot 2 depicted as consisting of four parcels containing a total of 153.98 acres and Lot l depicted as two parcels containing 160 acres (see Figure 10.11*a*). However, after a dependent resurvey and a subsequent retracement, it was determined that the closing lines were short, and the prorating of the distance of the two lots indicated that Lot 1 actually contained 156.24 acres and Lot 2 contained 150.6 acres. The question that the retracing surveyor must address is: What do I survey? Lot 1 or 2 or the area? It depends. Even though the original government surveyor did not physically run the boundary line between Lots 1 and 2, they were fixed legally, and the bona fide rights were established in accordance with the original survey. Accepting the fact that, although there are fewer acres present within the actual boundaries of Lots 1 and 2 than were indicated in the original survey, the answer would be: "First come gets the total area and the second gets the remainder." Thus, the bona fide rights of one would be jeopardized. The correct answer in locating the original boundary between Lots 1 and 2 would be to apply the principle that each lot takes its proportionate share of the deficiency, and the dividing line between the two retraced lots would be proportionate to the original. In this situation, the retracing surveyor must examine prior deeds, the chain of title, and possible unwritten rights.



(*a*)

Figure 10.11a Comparison of survey lines and areas.

There has been considerable manipulation with the federal method of describing land parcels. Federal laws permit legal subdivisions to a ¼ ¼ or 40 nominal acres. One can find many descriptions that will describe a parcel to 1 ¼ acres by subdivision. The law does not permit it. Also those who write descriptions play “loose” describing less than 40 acres. In some states descriptions such as NE ½ of the NE 1/4. This description, when challenged in the courts has been held valid in some and void in others.

Over the years, confusion and ambiguity have resulted from improper identification of portions of sections. A description that conveys the east 80 acres of the northeast quarter of a section does not convey the eastern half of the northeast quarter. The official plat of the section usually indicates a return of 640 acres, with all the lines 80 or 40 chains in the quarter section. Theoretically, this would place 160 acres in the northeast quarter and 80 acres in the eastern half of the quarter section. Since in a resurvey a section is rarely found to be the size reported, a description conveying the east 80 acres may or may not convey 80 acres for the following reasons.

If in the retracement of a section the surveyor finds that the section measures exactly the same as the record—an unlikely event—the east 80 acres and the eastern half of the northeast quarter would be identical. If in the retracement the section was found to be long in measurements and the northeast quarter contained more than 160 acres, only 80 acres would be conveyed. If the original patentee patented the eastern half, the seller of the east 80 acres would retain a strip of land. On the other hand, if the section measured short in size and the patentee had obtained only the eastern half, sale of the east 80 acres would not be a sale of 80 acres, since the seller did not own 80 acres. Persons describing land should refrain from conveying an area if it is intended that an aliquot portion be conveyed.

10.53 ACCORDING TO THE GOVERNMENT MEASURE

The words "according to the government measure" or "according to the government survey" become controlling elements in a description. When these words are discovered, the retracing surveyor must accept the realization that all the elements and the principles of retracing and locating a U.S. government parcel must be applied. That also requires that the retracing surveyor relate back to the original measurements that are indicated on the original plat and in the original notes. If the phrase "according to the government measure" is included as a portion of a description, the interpretation resorts to the original measure as indicated on the official plat, and the conveyance is intended to be a proportionate part of the original measure. With the use of these terms, the retraceemnt takes on an entirely new approach, in that the retracing surveyor should refer to as many of the original documents as possible. The insertion of these words give any description an entirely new meaning. They may become the controlling factor if questioned.

DIFFERENCES BETWEEN STATE AND FEDERAL INTERPRETATIONS

10.54 APPLYING STATE LAWS

It is an accepted principle that, once lands were divested from the United States into private ownership, the responsibility of retracement became that of the states. As long as the federal government retained any rights in any lands, the federal rules of survey must apply. Since the public surveys were completed, many states have enacted statutes that conflict with federal laws. If the state has final authority in a jurisdiction, the state law would be followed so far as property rights are concerned. Some of the deviations from federal laws are discussed in the following sections.

Missouri Statute Law

Missouri statute law (Section 60-290) provided that a lost section corner should be reestablished at the point of intersection of straight lines connecting the nearest corner found on each side of the lost corner; double proportionate measurement is not used. In the case of *Simpson* v. *Steward*, the court observed that double proportioning was one method of restoring a lost corner and the state rule was another.[17] As neither rule would restore a corner exactly where the original was, either rule would be equally valid, and in Missouri the state statute is enforced. Recently, this statute was repealed. Court opinion will be needed to clarify the status of the old and new situations.

Wisconsin Law

In 1862, a law was passed by the state of Wisconsin which provided that whenever a surveyor is required to make a subdivision of a section, as determined by the United States Survey, except where the section is fractional, the surveyor should establish the interior quarter-section corner at a point that is the same distance from the east quarter-section corner as it is from the west quarter-section corner, and the same distance from the north quarter-section corner as it is from the south quarter-section corner.[18] In 1867, the law was repealed, and new legislation provided that whenever a surveyor is required to subdivide a section or smaller subdivision of land established by the United States Survey, the surveyor should proceed according to the statutes of the United States and the rules and regulations made by the secretary of the interior in conformity thereto. In Wisconsin, the problem of the center of the section is now resolved by statute law.

10.55 TOPOGRAPHY

For the most part, the identification of topographic calls was one of the responsibilities of the original surveyors. There are times when the topographic calls harmonize well with actual conditions, and there are times when the retracing surveyor questions whether the original surveyor was even in the state, let alone the township, in that no harmonious relationship can be made of the results. The retracing surveyor will find mountains where a stream is indicated on the original map, or the retracing surveyor is positive that the original surveyor was there, in that all of the features check but are out of position by 20 chains. In such a situation, the retracing surveyor is presented with a major problem to ascertain just what happened.

The presumption is that the survey was conducted as it is indicated in the field notes and depicted on the plat. Since no court has the authority to determine that the survey was erroneous, the retracing surveyor can only report the facts. In California, if topography calls can be identified, the courts generally accept them, even though they may be substantially off distance calls. Reestablishing a corner by proportionate methods is truly a rule of last resort.

According to *Hanes* v. *Hollow Tree Lumber Co.*, "The government survey cannot be impeached by proof of fraud, negligence, or mistake on the part of the Government surveyor."[19] In a California court, as in some other states, it is never wise to say that the original survey was fraudulent; the surveyor should merely point out that topography calls and other items are not in agreement with the field notes.

10.56 BOUNDARIES BY AREA

Surveyors retracing boundaries from descriptions that refer to area may be confused as to the proper methods that were used to create the original boundaries. The surveyor will find that two basic approaches are accepted; which one should be used cannot be stated with certainty in a book such as this. A description of the "north ½ of the SE ¼" may be approached in several ways. In applying the federal approach, the north half is determined by taking one-half of the distance of all lines involved. This division may or may not give each divided portion an equal area. Under the state approach, the dividing lines are determined using equal area as the factor for locating the boundary lines. Not all states use this method; many of the GLO states use the federal method. It is important that a surveyor understand what method is used. In a California case, a description was written for Section 30. The original plat indicated the east half as containing 80 acres and the west half as containing 98.98 acres. A single patent was issued by the United States for the entire southwest quarter. Subsequently, the owner sold the east half. The buyer took possession of approximately 90 acres, or half of the total area that was based on a survey, not on the 80 acres depicted on the plat. In the ensuing litigation the court awarded half of the area, noting that the owner could sell the parcel by any description that he or she wanted to use. This decision probably would have been different had the deed made reference to "according to the government survey."

This should be a warning to surveyors who are asked to identify or survey boundaries of such parcels. Aliquot portions should be surveyed according to the federal rules of survey, but after lands have been patented into private ownership, the state rules may apply and equal area will be considered, unless the magic words "according to the government survey" are recited. In the opinion cited here, if the United States had patented the two parcels to two separate persons, the decision would have been different. In that specific case, the federal rules would have applied.

By the federal aliquot part rule, one-half is seldom one-half of the area of a whole; by state common law, one-half is one-half the area of a whole. In *Wood* v. *Mandrilla*,[20] in Section 30, T 20 S., R 24 E. Mount Diablo Meridian, the original township plat pictured the east half as 80 acres and the west half as 98.98 acres. One patent was issued for the entire southwest quarter, and the owner sold the east half. The buyer took possession of half the area, about 90 acres, instead of the 80 acres shown on the government plat.

According to the court, the seller has the right to sell by either the federal method or by proportion of the area; the court awarded half the area. Aliquot parcels obtained by patent from the federal government should be surveyed in accordance with federal rules, but after a patent has been issued and the land is divided by fractional parts, the state rule of equal area may apply. In the *Wood* v. *Mandrilla* case cited here, if the east half had been patented from the federal government separately from the west half, federal rules would apply.

10.57 ESTABLISHING CORNERS

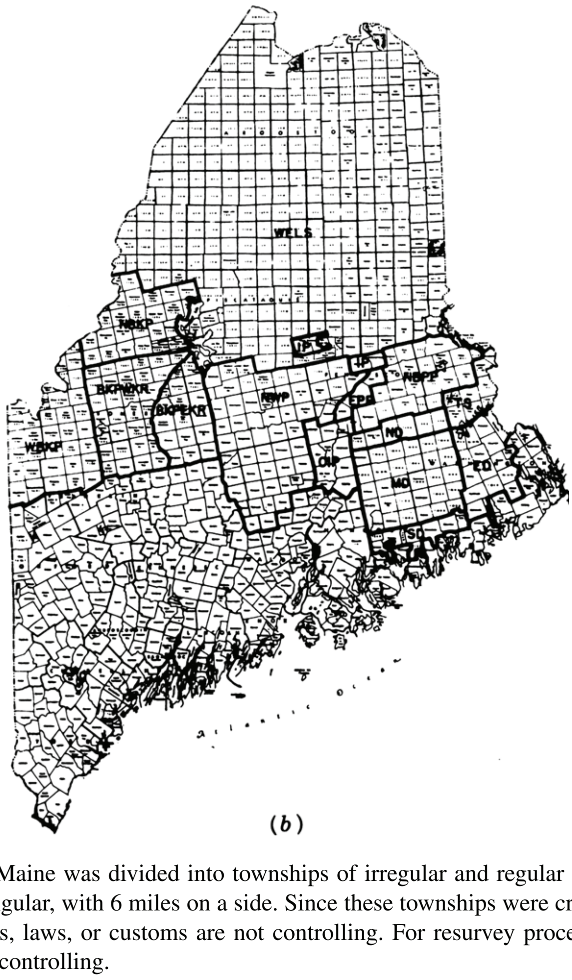
A few states have passed laws whereby the county surveyor could "establish" a corner or boundary line by following the procedure required by law. If the county surveyor's findings were not appealed within a definite period of time, the findings became final. Very often, the county surveyor located centers of sections or sixteenth corners in violation of the law. When it could be shown that the center of the section was not on the intersection of lines connecting opposite quarter corners, the question arose: Is the corner located by the county surveyor the legal center of the section, or should the true center be located? In general, where the county surveyor set the corner in accordance with statutory requirements and the period of time stated in the statute has passed, the county surveyor's corner becomes the legal center of the section for parties properly notified. Such proceedings are much like a court case.

Unless it can be proven that legal process was not followed, no future survey can supersede the one established by law.

10.58 SECTIONS CREATED UNDER STATE JURISDICTION

Not all sectionalized lands were surveyed and created under federal laws (see Figure 10.11*b*).

In West Texas, many sections were originally surveyed under the authority of Texas; these lands were never owned by the federal government. This also applies to a major portion of Northern Maine and throughout Northeastern U.S. In California, much confusion was created when a law was passed making it necessary for the assessor to assess parcels not greater than 1 square mile. Most of the owners of



(*b*)

Figure 10.11b Maine was divided into townships of irregular and regular shape. Later surveys were rectangular, with 6 miles on a side. Since these townships were created under state law, federal rules, laws, or customs are not controlling. For resurvey procedure, state court interpretation is controlling.

large ranchos filed maps projecting section lines through the rancho as they would have existed if sections had been surveyed. Later, lands were sold by the theoretical sections (never monumented). In a few instances, the sections were monumented by variable methods, depending on the surveyor.

If the sections of land were not created by federal rules of the Bureau of Land Management, or its predecessors, and were not under federal jurisdiction, the foregoing principles are not necessarily applicable. State laws must be examined to find proper procedures. In keeping with the majority opinions and rules now accepted by many states, state courts, and knowledgeable surveyors, the federal rules for subdividing sections of land that were originally created under the federal laws apply when retracing those lands:

> The description of the original parcel may determine which law(s) to apply in a retracement.[21]

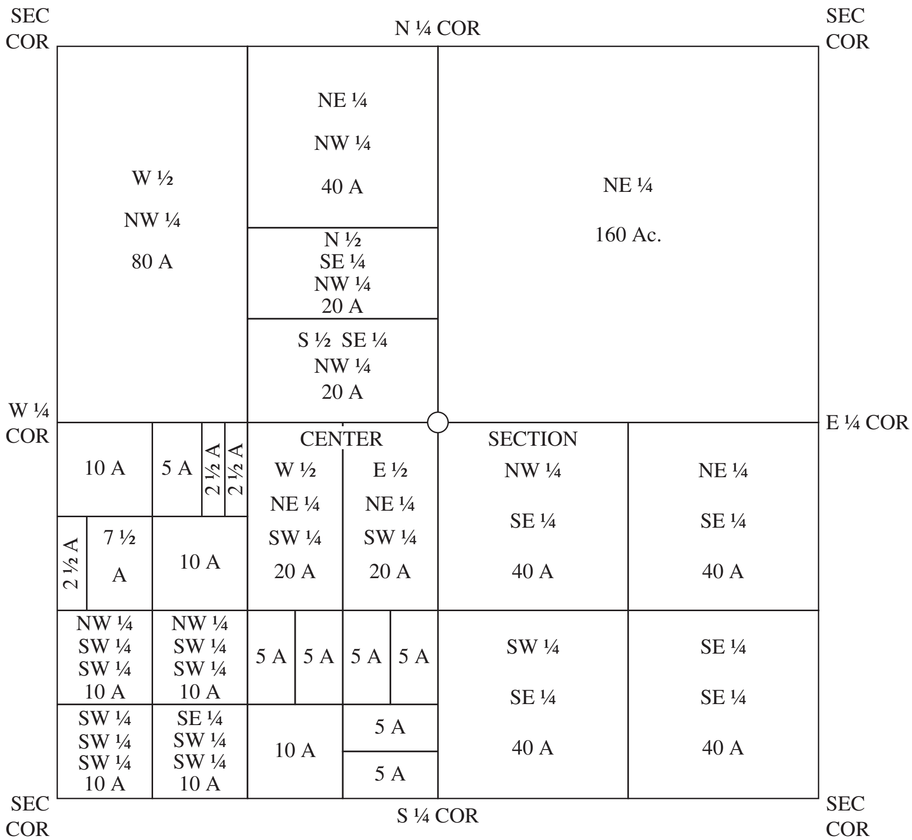
10.59 PRESUMPTIONS AND REALITIES FOR GLO SURVEYS

A surveyor retracing the lines of aliquot portions of GLO surveys not only should have an understanding of the theoretical township or sections but should also understand the actual or realistic township and sections. The following facts may apply both to the theoretical sections and to those as actually surveyed (Figure 10.12). In an actual presumed, theoretical, legal, statutory section that was anticipated by the original surveys, we should anticipate the following:

1. All bearings are recited in the notes and depicted on the plats as north, south, east, and west, or otherwise as indicated.
2. All sides are 80 chains in length and not 5280 feet or as indicated in the notes and depicted on the plats.
3. All other distances indicated are the true surveyed field distances.
4. All angles at the corners are right angles.
5. Each section contains the exact number of acres indicated.
6. Quarter corners are equidistant between section corners.
7. Quarter corners are on a straight line between section corners.
8. The legal center of the section is the geometric center.
9. Opposite sides of the section are parallel.
10. The section contains exactly 640 acres, and the aliquot parts contain the exact aliquot share of the entire section.

The true section found as a result of a resurvey or retracement is as follows (Figure 10.12):

1. The sides of the section are not 80 chains long.
2. Quarter-section sides are not 40 chains long.



SECTION OF LAND (640 A)
WITH THEORETICAL SUBDIVISIONS

Figure 10.12 Idealized interior section. Shown are standard aliquot part designations (with nominal areas) within an interior section, according to the U.S. Public Lands Survey System. Those sections against the north and west boundaries of a township, those quarter-quarters (or "forties") which occur against those north and west boundaries are designated as lots and not aliquot pans of sections, in addition, other specialized situations may also require that certain parcels of land have lot or tract designations. For sections that more nearly resemble real-world conditions, see Figure 10.13.

3. Quarter-section corners are not equidistant between section corners.
4. Quarter-section corners are not on a straight line between section corners.
5. No one line is on a true cardinal bearing.
6. The legal center of the section is not the geometric center.
7. No angle between lines is 90°.

TYPICAL SECTION AS LAID OUT

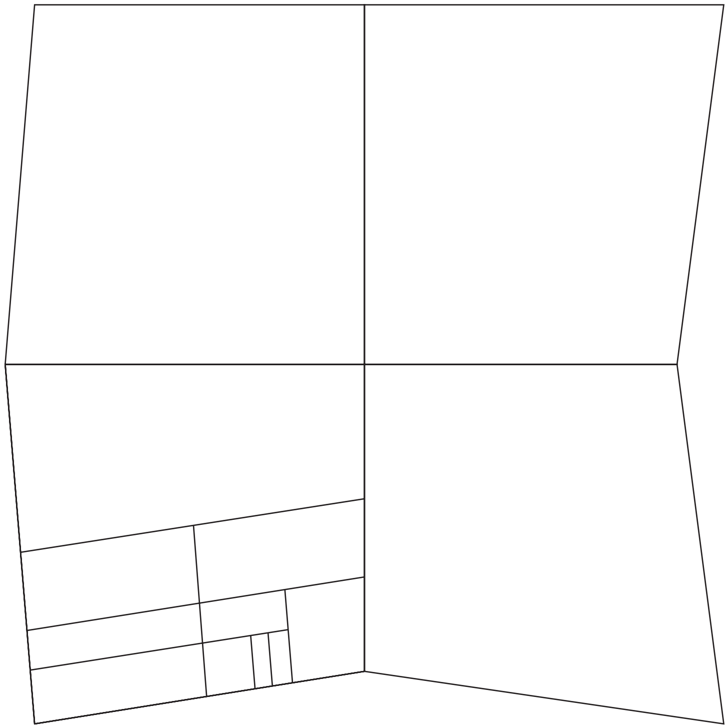


Figure 10.13 Typical "regular" section. *Each* section must be considered a *special* case.

8. Opposite sides of the section are not parallel.
9. The section does not contain exactly 640 acres, nor do any of the subdivision parts contain their exact theoretical acreage.
10. The actual field measured distances and bearings do not agree with those reported originally.

10.60 CONCLUSIONS

Principle 24. *Other than creating new subdivisions for private landowners, today the majority of the modern surveyor's work is retracing the lines of the original surveys, some of which are ancient in nature.*

Today, the majority of the services performed by land surveyors is finding previously run parcels of land. Granted, some original surveys are conducted for the creation of new subdivisions, but this does not constitute the major portion of a surveyor's work. Most surveyors conduct retracements of previously created parcels of land, both federally created parcels and private parcels. Few surveyors concentrate on creating new land parcels or townships. The federal government conducts few original surveys in the GLO states—Alaska being the exception. Some surveyors' entire work is running previously created lines and then redescribing them for today's terms for today's landowners. A few firms may still prepare new subdivisions for the sale of the newly created lots or parcels, but in many areas this is the exception. Since the future of the land surveyor may be in the area of retracements, it is imperative that the modern surveyor have a complete understanding of the differences between the methods of retracement and an original survey.

NOTES

1. *Manual for the survey of the Public Lands of the United States, Bureau of Land MAnagement, U.S. D,I. printed by the Government Printing Office, 2009.*
2. *Cragin* v. *Powell*, 128 U.S. 691, 2 S.Ct. 203 (La. 1888).
3. U.S.C.A., Title 43, Sec. 472.
4. Supra note 1.
5. *Harrington* v. *Boehamer*, 134 Cal. 196 (1901).
6. Bureau of Land Management, *Manual of Instructions for the Survey of Public Lands* (Washington, DC: U.S. Department of the Interior, 1973), 310.
7. Pulehunui, 4 H 239 (1879).
8. *U.S.* v. *John Citko et ux*, 517 F.Supp. 233 (1981).
9. 468 F.2d 633 (1972).
10. *Restoration of Lost or Obliterated Corners* (Washington, DC: U.S. Government Printing Office, 1976), 6.
11. Ibid., 13.
12. *Manual*, 366.
13. U.S.C.A., Title 43, Sec. 752.
14. Ibid.
15. Ibid.
16. *Lee Wilson & Co*. v. *U.S*., 245 U.S. 24 (1917).
17. *Simpson* v. *Steward*, 281 Mo. 228 (1919).
18. Wisc. Stat Ch. 120, Sec. 4.
19. *Hanes* v. *Hollow Tree Lumber Co*., 191 C.A.2d 658 (1961).
20. *Wood* v. *Mandrilla*, 167 C.A. 607 (1914).
21. *Bryant* v. *Blevins*, 884 P2d 1034 (Cal. 1994).